

SIGNALS AND SYSTEMS

In this chapter we shall discuss basic aspects of signals and systems. We shall also introduce fundamental concepts and qualitative explanations of the hows and whys of systems theory, thus building a solid foundation for understanding the quantitative analysis in the remainder of the book. For simplicity, the focus of this chapter is on continuous-time signals and systems. Chapter 3 presents the same ideas for discrete-time signals and systems.

SIGNALS

A *signal* is a set of data or information. Examples include a telephone or a television signal, monthly sales of a corporation, or daily closing prices of a stock market (e.g., the Dow Jones averages). In all these examples, the signals are functions of the independent variable *time*. This is not always the case, however. When an electrical charge is distributed over a body, for instance, the signal is the charge density, a function of *space* rather than time. In this book we deal almost exclusively with signals that are functions of time. The discussion, however, applies equally well to other independent variables.

SYSTEMS

Signals may be processed further by *systems*, which may modify them or extract additional information from them. For example, an anti-aircraft gun operator may want to know the future location of a hostile moving target that is being tracked by his radar. Knowing the radar signal, he knows the past location and velocity of the target. By properly processing the radar signal (the input), he can approximately estimate the future location of the target. Thus, a system is an entity that *processes* a set of signals (*inputs*) to yield another set of signals (*outputs*). A system may be made up of physical components, as in electrical, mechanical, or hydraulic systems (hardware realization), or it may be an algorithm that computes an output from an input signal (software realization).

1.1 SIZE OF A SIGNAL

The size of any entity is a number that indicates the largeness or strength of that entity. Generally speaking, the signal amplitude varies with time. How can a signal that exists over a certain time

interval with varying amplitude be measured by one number that will indicate the signal size or signal strength? Such a measure must consider not only the signal amplitude, but also its duration. For instance, if we are to devise a single number V as a measure of the size of a human being, we must consider not only his or her width (girth), but also the height. If we make a simplifying assumption that the shape of a person is a cylinder of variable radius r (which varies with the height h), then one possible measure of the size of a person of height H is the person's volume V , given by

$$V = \pi \int_0^H r^2(h) dh$$

1.1-1 Signal Energy

Arguing in this manner, we may consider the area under a signal $x(t)$ as a possible measure of its size, because it takes account not only of the amplitude but also of the duration. However, this will be a defective measure because even for a large signal $x(t)$, its positive and negative areas could cancel each other, indicating a signal of small size. This difficulty can be corrected by defining the signal size as the area under $|x(t)|^2$, which is always positive. We call this measure the *signal energy* E_x , defined as

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad (1.1)$$

This definition simplifies for a real-valued signal $x(t)$ to $E_x = \int_{-\infty}^{\infty} x^2(t) dt$. There are also other possible measures of signal size, such as the area under $|x(t)|$. The energy measure, however, is not only more tractable mathematically but is also more meaningful (as shown later) in the sense that it is indicative of the energy that can be extracted from the signal.

1.1-2 Signal Power

Signal energy must be finite for it to be a meaningful measure of signal size. A necessary condition for the energy to be finite is that the signal amplitude $\rightarrow 0$ as $|t| \rightarrow \infty$ (Fig. 1.1a). Otherwise the integral in Eq. (1.1) will not converge.

When the amplitude of $x(t)$ does not $\rightarrow 0$ as $|t| \rightarrow \infty$ (Fig. 1.1b), the signal energy is infinite. A more meaningful measure of the signal size in such a case would be the time average of the energy, if it exists. This measure is called the *power* of the signal. For a signal $x(t)$, we define its power P_x as

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \quad (1.2)$$

This definition simplifies for a real-valued signal $x(t)$ to $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$. Observe that the signal power P_x is the time average (mean) of the signal magnitude squared, that is, the *mean-square* value of $|x(t)|$. Indeed, the square root of P_x is the familiar *rms* (root-mean-square) value of $x(t)$.

Generally, the mean of an entity averaged over a large time interval approaching infinity exists if the entity either is periodic or has a statistical regularity. If such a condition is not satisfied, the average may not exist. For instance, a ramp signal $x(t) = t$ increases indefinitely as $|t| \rightarrow \infty$, and neither the energy nor the power exists for this signal. However, the unit step function, which is not periodic nor has statistical regularity, does have a finite power.

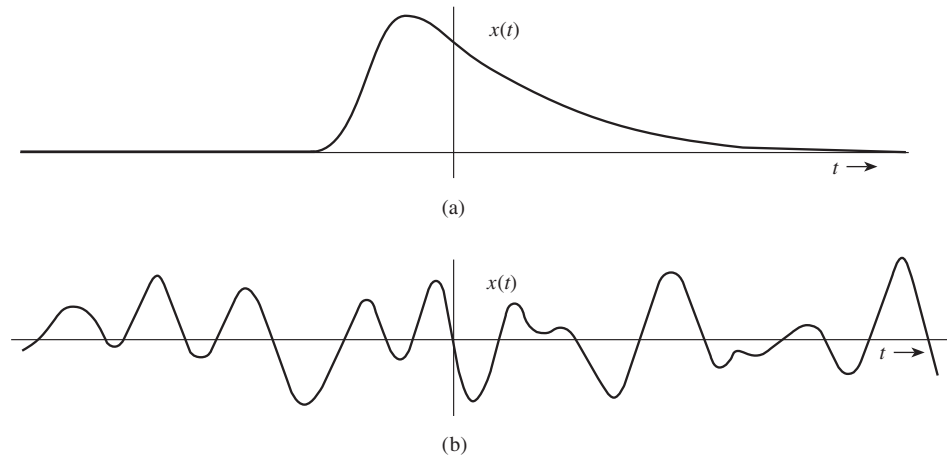


Figure 1.1 Examples of signals: **(a)** a signal with finite energy and **(b)** a signal with finite power.

When $x(t)$ is periodic, $|x(t)|^2$ is also periodic. Hence, the power of $x(t)$ can be computed from Eq. (1.2) by averaging $|x(t)|^2$ over one period.

Comments. The signal energy as defined in Eq. (1.1) does not indicate the actual energy (in the conventional sense) of the signal because the signal energy depends not only on the signal, but also on the load. It can, however, be interpreted as the energy dissipated in a normalized load of a 1 ohm resistor if a voltage $x(t)$ were to be applied across the 1 ohm resistor [or if a current $x(t)$ were to be passed through the 1 ohm resistor]. The measure of “energy” is therefore indicative of the energy capability of the signal, not the actual energy. For this reason the concepts of conservation of energy should not be applied to this “signal energy.” Parallel observation applies to “signal power” defined in Eq. (1.2). These measures are but convenient indicators of the signal size, which prove useful in many applications. For instance, if we approximate a signal $x(t)$ by another signal $g(t)$, the error in the approximation is $e(t) = x(t) - g(t)$. The energy (or power) of $e(t)$ is a convenient indicator of the goodness of the approximation. It provides us with a quantitative measure of determining the closeness of the approximation. In communication systems, during transmission over a channel, message signals are corrupted by unwanted signals (noise). The quality of the received signal is judged by the relative sizes of the desired signal and the unwanted signal (noise). In this case the ratio of the message signal and noise signal powers (signal-to-noise power ratio) is a good indication of the received signal quality.

Units of Energy and Power. Equation (1.1) is not correct dimensionally. This is because here we are using the term *energy* not in its conventional sense, but to indicate the signal size. The same observation applies to Eq. (1.2) for power. The units of energy and power, as defined here, depend on the nature of the signal $x(t)$. If $x(t)$ is a voltage signal, its energy E_x has units of volts squared-seconds ($V^2 s$), and its power P_x has units of volts squared. If $x(t)$ is a current signal, these units will be amperes squared-seconds ($A^2 s$) and amperes squared, respectively.

EXAMPLE 1.1 Classifying Energy and Power Signals

Determine the suitable measures of the signals in Fig. 1.2.

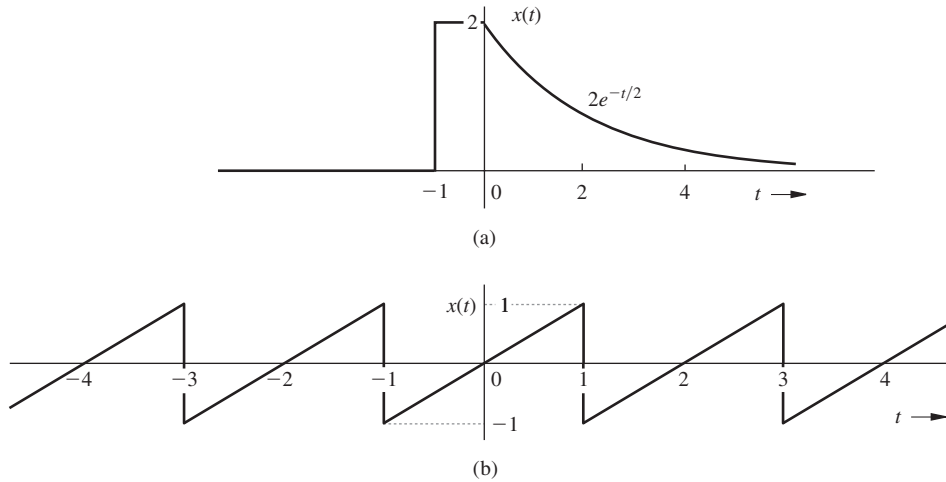


Figure 1.2 Signals for Ex. 1.1

In Fig. 1.2a, the signal amplitude $\rightarrow 0$ as $|t| \rightarrow \infty$. Therefore the suitable measure for this signal is its energy E_x given by

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-1}^0 (2)^2 dt + \int_0^{\infty} 4e^{-t} dt = 4 + 4 = 8$$

In Fig. 1.2b, the signal magnitude does not $\rightarrow 0$ as $|t| \rightarrow \infty$. However, it is periodic, and therefore its power exists. We can use Eq. (1.2) to determine its power. We can simplify the procedure for periodic signals by observing that a periodic signal repeats regularly each period (2 seconds in this case). Therefore, averaging $|x(t)|^2$ over an infinitely large interval is identical to averaging this quantity over one period (2 seconds in this case). Thus

$$P_x = \frac{1}{2} \int_{-1}^1 |x(t)|^2 dt = \frac{1}{2} \int_{-1}^1 t^2 dt = \frac{1}{3}$$

Recall that the signal power is the square of its rms value. Therefore, the rms value of this signal is $1/\sqrt{3}$.

EXAMPLE 1.2 Determining Power and RMS Value

Determine the power and the rms value of

- (a) $x(t) = C \cos(\omega_0 t + \theta)$
- (b) $x(t) = C_1 \cos(\omega_1 t + \theta_1) + C_2 \cos(\omega_2 t + \theta_2) \quad \omega_1 \neq \omega_2$
- (c) $x(t) = D e^{j\omega_0 t}$

(a) This is a periodic signal with period $T_0 = 2\pi/\omega_0$. The suitable measure of this signal is its power. Because it is a periodic signal, we may compute its power by averaging its energy over one period $T_0 = 2\pi/\omega_0$. However, for the sake of demonstration, we shall use Eq. (1.2) to solve this problem by averaging over an infinitely large time interval.

$$\begin{aligned} P_x &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} C^2 \cos^2(\omega_0 t + \theta) dt = \lim_{T \rightarrow \infty} \frac{C^2}{2T} \int_{-T/2}^{T/2} [1 + \cos(2\omega_0 t + 2\theta)] dt \\ &= \lim_{T \rightarrow \infty} \frac{C^2}{2T} \int_{-T/2}^{T/2} dt + \lim_{T \rightarrow \infty} \frac{C^2}{2T} \int_{-T/2}^{T/2} \cos(2\omega_0 t + 2\theta) dt \end{aligned}$$

The first term on the right-hand side is equal to $C^2/2$. The second term, however, is zero because the integral appearing in this term represents the area under a sinusoid over a very large time interval T with $T \rightarrow \infty$. This area is at most equal to the area of half the cycle because of cancellations of the positive and negative areas of a sinusoid. The second term is this area multiplied by $C^2/2T$ with $T \rightarrow \infty$. Clearly this term is zero, and

$$P_x = \frac{C^2}{2}$$

This shows that a sinusoid of amplitude C has a power $C^2/2$ regardless of the value of its frequency ω_0 ($\omega_0 \neq 0$) and phase θ . The rms value is $C/\sqrt{2}$. If the signal frequency is zero (dc or a constant signal of amplitude C), the reader can show that the power is C^2 .

(b) In Ch. 6, we shall show that a sum of two sinusoids may or may not be periodic, depending on whether the ratio ω_1/ω_2 is a rational number. Therefore, the period of this signal is not known. Hence, its power will be determined by averaging its energy over T seconds with $T \rightarrow \infty$. Thus,

$$\begin{aligned} P_x &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [C_1 \cos(\omega_1 t + \theta_1) + C_2 \cos(\omega_2 t + \theta_2)]^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} C_1^2 \cos^2(\omega_1 t + \theta_1) dt + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} C_2^2 \cos^2(\omega_2 t + \theta_2) dt \\ &\quad + \lim_{T \rightarrow \infty} \frac{2C_1 C_2}{T} \int_{-T/2}^{T/2} \cos(\omega_1 t + \theta_1) \cos(\omega_2 t + \theta_2) dt \end{aligned}$$

The first and second integrals on the right-hand side are the powers of the two sinusoids, which are $C_1^2/2$ and $C_2^2/2$, as found in part (a). The third term, the product of two sinusoids, can be expressed as a sum of two sinusoids $\cos[(\omega_1 + \omega_2)t + (\theta_1 + \theta_2)]$ and $\cos[(\omega_1 - \omega_2)t + (\theta_1 - \theta_2)]$, respectively. Now, arguing as in part (a), we see that the third term is zero. Hence, we have[†]

$$P_x = \frac{C_1^2}{2} + \frac{C_2^2}{2}$$

and the rms value is $\sqrt{(C_1^2 + C_2^2)/2}$.

We can readily extend this result to a sum of any number of sinusoids with distinct frequencies. Thus, if

$$x(t) = \sum_{n=1}^{\infty} C_n \cos(\omega_n t + \theta_n)$$

assuming that none of the two sinusoids have identical frequencies and $\omega_n \neq 0$, then

$$P_x = \frac{1}{2} \sum_{n=1}^{\infty} C_n^2$$

If $x(t)$ also has a dc term, as

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(\omega_n t + \theta_n)$$

then

$$P_x = C_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} C_n^2 \quad (1.3)$$

(c) In this case the signal is complex, and we use Eq. (1.2) to compute the power.

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |De^{j\omega_0 t}|^2 dt$$

Recall that $|e^{j\omega_0 t}| = 1$ so that $|De^{j\omega_0 t}|^2 = |D|^2$, and

$$P_x = |D|^2 \quad (1.4)$$

The rms value is $|D|$.

Comment. In part (b) of Ex. 1.2, we have shown that the power of the sum of two sinusoids is equal to the sum of the powers of the sinusoids. It may appear that the power of $x_1(t) + x_2(t)$

[†] This is true only if $\omega_1 \neq \omega_2$. If $\omega_1 = \omega_2$, the integrand of the third term contains a constant $\cos(\theta_1 - \theta_2)$, and the third term $\rightarrow 2C_1C_2 \cos(\theta_1 - \theta_2)$ as $T \rightarrow \infty$.

is $P_{x_1} + P_{x_2}$. Unfortunately, this conclusion is not true in general. It is true only under a certain condition (orthogonality), discussed later (Sec. 6.5-3).

DRILL 1.1 Computing Energy, Power, and RMS Value

Show that the energies of the signals in Figs. 1.3a, 1.3b, 1.3c, and 1.3d are 4, 1, $4/3$, and $4/3$, respectively. Observe that doubling a signal quadruples the energy, and time-shifting a signal has no effect on the energy. Show also that the power of the signal in Fig. 1.3e is 0.4323. What is the rms value of signal in Fig. 1.3e?

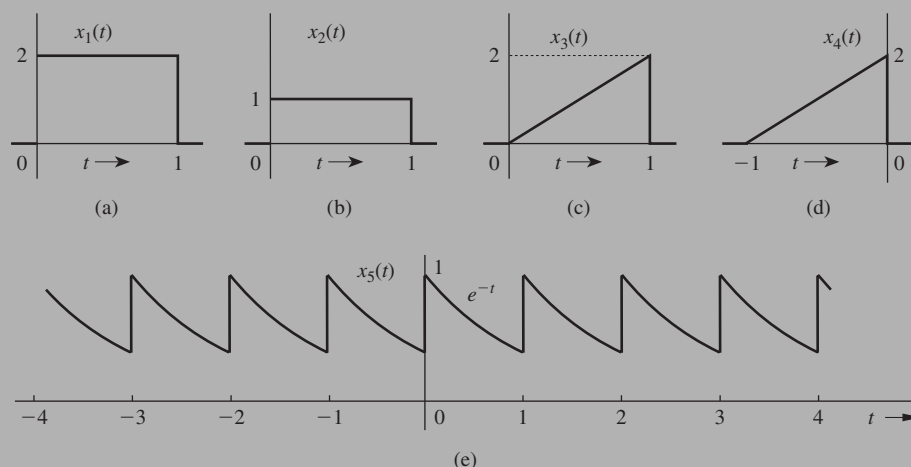


Figure 1.3 Signals for Drill 1.1

DRILL 1.2 Computing Power over a Period

Redo Ex. 1.1a to find the power of a sinusoid $C \cos(\omega_0 t + \theta)$ by averaging the signal energy over one period $T_0 = 2\pi/\omega_0$ (rather than averaging over the infinitely large interval). Show also that the power of a dc signal $x(t) = C_0$ is C_0^2 , and its rms value is C_0 .

DRILL 1.3 Power of a Sum of Two Equal-Frequency Sinusoids

Show that if $\omega_1 = \omega_2$, the power of $x(t) = C_1 \cos(\omega_1 t + \theta_1) + C_2 \cos(\omega_2 t + \theta_2)$ is $[C_1^2 + C_2^2 + 2C_1 C_2 \cos(\theta_1 - \theta_2)]/2$, which is not equal to the Ex. 1.2b result of $(C_1^2 + C_2^2)/2$.

1.2 SOME USEFUL SIGNAL OPERATIONS

We discuss here three useful signal operations: shifting, scaling, and inversion. Since the independent variable in our signal description is time, these operations are discussed as *time shifting*, *time scaling*, and *time reversal* (inversion). However, this discussion is valid for functions having independent variables other than time (e.g., frequency or distance).

1.2-1 Time Shifting

Consider a signal $x(t)$ (Fig. 1.4a) and the same signal delayed by T seconds (Fig. 1.4b), which we shall denote by $\phi(t)$. Whatever happens in $x(t)$ (Fig. 1.4a) at some instant t also happens in $\phi(t)$ (Fig. 1.4b) T seconds later at the instant $t + T$. Therefore

$$\phi(t + T) = x(t) \quad \text{and} \quad \phi(t) = x(t - T)$$

Therefore, to time-shift a signal by T , we replace t with $t - T$. Thus $x(t - T)$ represents $x(t)$ time-shifted by T seconds. If T is positive, the shift is to the right (delay), as in Fig. 1.4b. If T is negative, the shift is to the left (advance), as in Fig. 1.4c. Clearly, $x(t - 2)$ is $x(t)$ delayed (right-shifted) by 2 seconds, and $x(t + 2)$ is $x(t)$ advanced (left-shifted) by 2 seconds.

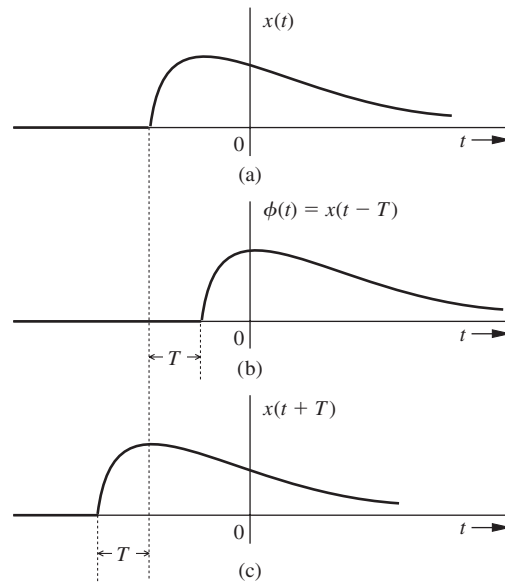


Figure 1.4 Time-shifting a signal.

EXAMPLE 1.3 Time Shifting

An exponential function $x(t) = e^{-2t}$ shown in Fig. 1.5a is delayed by 1 second. Sketch and mathematically describe the delayed function. Repeat the problem with $x(t)$ advanced by 1 second.

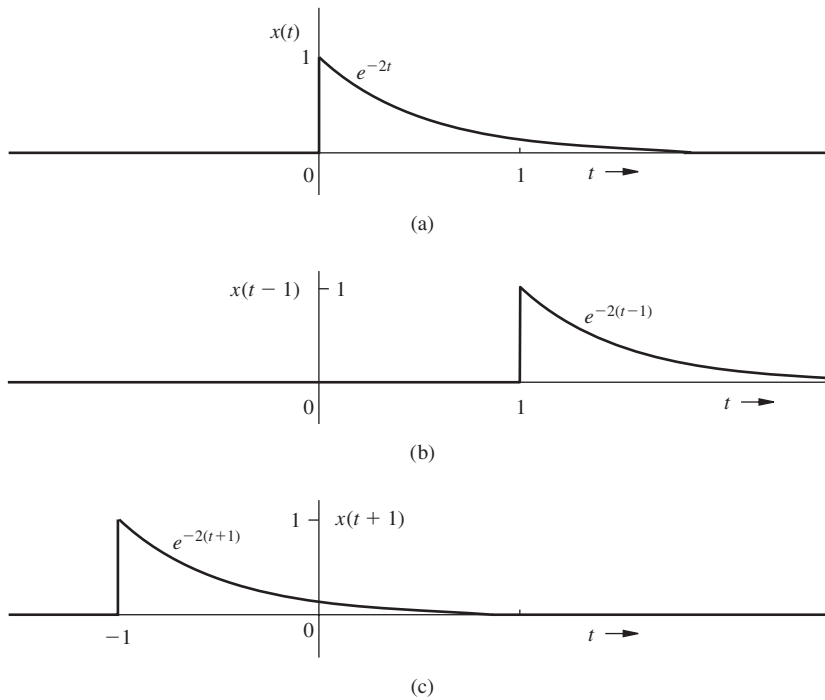


Figure 1.5 (a) Signal $x(t)$. (b) Signal $x(t)$ delayed by 1 second. (c) Signal $x(t)$ advanced by 1 second.

The function $x(t)$ can be described mathematically as

$$x(t) = \begin{cases} e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (1.5)$$

Let $x_d(t)$ represent the function $x(t)$ delayed (right-shifted) by 1 second, as illustrated in Fig. 1.5b. This function is $x(t-1)$; its mathematical description can be obtained from $x(t)$

by replacing t with $t - 1$ in Eq. (1.5). Thus,

$$x_d(t) = x(t - 1) = \begin{cases} e^{-2(t-1)} & t - 1 \geq 0 \quad \text{or} \quad t \geq 1 \\ 0 & t - 1 < 0 \quad \text{or} \quad t < 1 \end{cases}$$

Let $x_a(t)$ represent the function $x(t)$ advanced (left-shifted) by 1 second, as depicted in Fig. 1.5c. This function is $x(t + 1)$; its mathematical description can be obtained from $x(t)$ by replacing t with $t + 1$ in Eq. (1.5). Thus,

$$x_a(t) = x(t + 1) = \begin{cases} e^{-2(t+1)} & t + 1 \geq 0 \quad \text{or} \quad t \geq -1 \\ 0 & t + 1 < 0 \quad \text{or} \quad t < -1 \end{cases}$$

DRILL 1.4 Working with Time Delay and Time Advance

Write a mathematical description of the signal $x_3(t)$ in Fig. 1.3c. Next, delay this signal by 2 seconds. Sketch the delayed signal. Show that this delayed signal $x_d(t)$ can be described mathematically as $x_d(t) = 2(t - 2)$ for $2 \leq t \leq 3$, and equal to 0 otherwise. Now repeat the procedure with the signal advanced (left-shifted) by 1 second. Show that this advanced signal $x_a(t)$ can be described as $x_a(t) = 2(t + 1)$ for $-1 \leq t \leq 0$, and 0 otherwise.

1.2-2 Time Scaling

The compression or expansion of a signal in time is known as *time scaling*. Consider the signal $x(t)$ of Fig. 1.6a. The signal $\phi(t)$ in Fig. 1.6b is $x(t)$ compressed in time by a factor of 2. Therefore, whatever happens in $x(t)$ at some instant t also happens to $\phi(t)$ at the instant $t/2$ so that

$$\phi\left(\frac{t}{2}\right) = x(t) \quad \text{and} \quad \phi(t) = x(2t)$$

Observe that because $x(t) = 0$ at $t = T_1$ and T_2 , we must have $\phi(t) = 0$ at $t = T_1/2$ and $T_2/2$, as shown in Fig. 1.6b. If $x(t)$ were recorded on a tape and played back at twice the normal recording speed, we would obtain $x(2t)$. In general, if $x(t)$ is compressed in time by a factor a ($a > 1$), the resulting signal $\phi(t)$ is given by

$$\phi(t) = x(at)$$

Using a similar argument, we can show that $x(t)$ expanded (slowed down) in time by a factor a ($a > 1$) is given by

$$\phi(t) = x\left(\frac{t}{a}\right)$$

Figure 1.6c shows $x(t/2)$, which is $x(t)$ expanded in time by a factor of 2. Observe that in a time-scaling operation, the origin $t = 0$ is the anchor point, which remains unchanged under the scaling operation because at $t = 0$, $x(t) = x(at) = x(0)$.

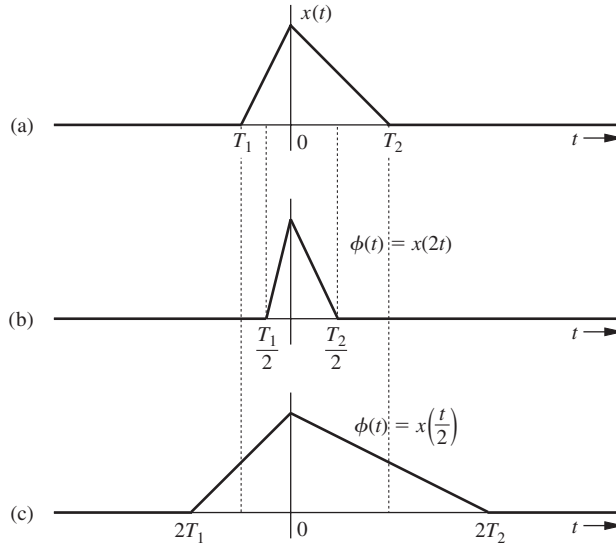


Figure 1.6 Time scaling a signal.

In summary, to time-scale a signal by a factor a , we replace t with at . If $a > 1$, the scaling results in compression, and if $a < 1$, the scaling results in expansion.

EXAMPLE 1.4 Continuous Time-Scaling Operation

Figure 1.7a shows a signal $x(t)$. Sketch and describe mathematically this signal time-compressed by factor 3. Repeat the problem for the same signal time-expanded by factor 2.

The signal $x(t)$ can be described as

$$x(t) = \begin{cases} 2 & -1.5 \leq t < 0 \\ 2e^{-t/2} & 0 \leq t < 3 \\ 0 & \text{otherwise} \end{cases} \quad (1.6)$$

Figure 1.7b shows $x_c(t)$, which is $x(t)$ time-compressed by factor 3; consequently, it can be described mathematically as $x(3t)$, which is obtained by replacing t with $3t$ in the right-hand side of Eq. (1.6). Thus,

$$x_c(t) = x(3t) = \begin{cases} 2 & -1.5 \leq 3t < 0 \quad \text{or} \quad -0.5 \leq t < 0 \\ 2e^{-3t/2} & 0 \leq 3t < 3 \quad \text{or} \quad 0 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

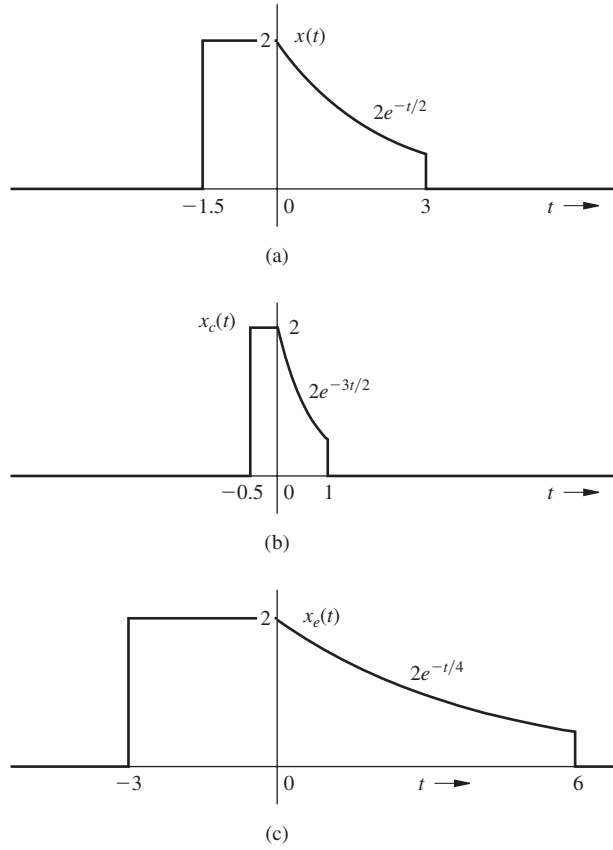


Figure 1.7 (a) Signal $x(t)$, (b) signal $x(3t)$, and (c) signal $x(t/2)$.

Observe that the instants $t = -1.5$ and 3 in $x(t)$ correspond to the instants $t = -0.5$, and 1 in the compressed signal $x(3t)$.

Figure 1.7c shows $x_e(t)$, which is $x(t)$ time-expanded by factor 2; consequently, it can be described mathematically as $x(t/2)$, which is obtained by replacing t with $t/2$ in $x(t)$. Thus,

$$x_e(t) = x\left(\frac{t}{2}\right) = \begin{cases} 2 & -1.5 \leq \frac{t}{2} < 0 \quad \text{or} \quad -3 \leq t < 0 \\ 2e^{-t/4} & 0 \leq \frac{t}{2} < 3 \quad \text{or} \quad 0 \leq t < 6 \\ 0 & \text{otherwise} \end{cases}$$

Observe that the instants $t = -1.5$ and 3 in $x(t)$ correspond to the instants $t = -3$ and 6 in the expanded signal $x(t/2)$.

DRILL 1.5 Compression and Expansion of Sinusoids

Show that the time compression by an integer factor n ($n > 1$) of a sinusoid results in a sinusoid of the same amplitude and phase, but with the frequency increased n -fold. Similarly, the time expansion by an integer factor n ($n > 1$) of a sinusoid results in a sinusoid of the same amplitude and phase, but with the frequency reduced by a factor n . Verify your conclusion by sketching a sinusoid $\sin 2t$ and the same sinusoid compressed by a factor 3 and expanded by a factor 2.

1.2-3 Time Reversal

Consider the signal $x(t)$ in Fig. 1.8a. We can view $x(t)$ as a rigid wire frame hinged at the vertical axis. To time-reverse $x(t)$, we rotate this frame 180° about the vertical axis. This time reversal [the reflection of $x(t)$ about the vertical axis] gives us the signal $\phi(t)$ (Fig. 1.8b). Observe that whatever happens in Fig. 1.8a at some instant t also happens in Fig. 1.8b at the instant $-t$, and vice versa. Therefore,

$$\phi(t) = x(-t)$$

Thus, to time-reverse a signal we replace t with $-t$, and the time reversal of signal $x(t)$ results in a signal $x(-t)$. We must remember that the reversal is performed about the vertical axis, which acts as an anchor or a hinge. Recall also that the reversal of $x(t)$ about the horizontal axis results in $-x(t)$.

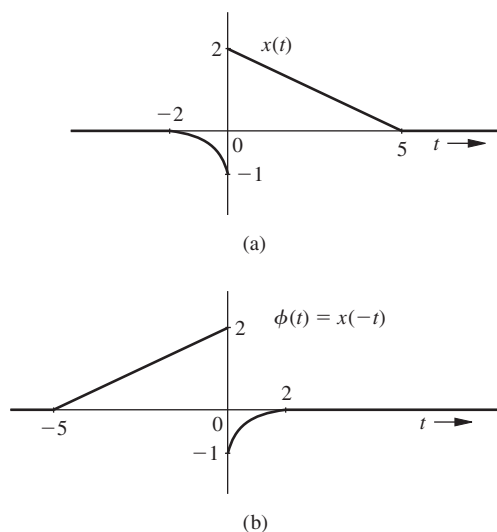


Figure 1.8 Time reversal of a signal.

EXAMPLE 1.5 Time Reversal of a Signal

For the signal $x(t)$ illustrated in Fig. 1.9a, sketch $x(-t)$, which is time-reversed $x(t)$.

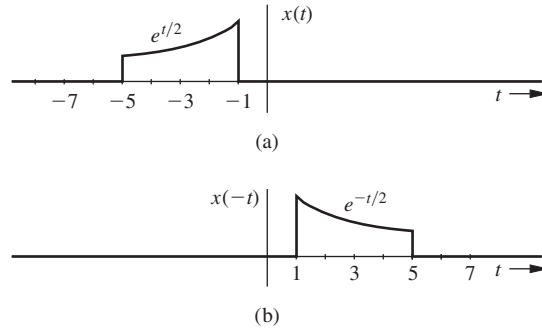


Figure 1.9 Example of time reversal.

The instants -1 and -5 in $x(t)$ are mapped into instants 1 and 5 in $x(-t)$. Because $x(t) = e^{t/2}$, we have $x(-t) = e^{-t/2}$. The signal $x(-t)$ is depicted in Fig. 1.9b. We can describe $x(t)$ and $x(-t)$ as

$$x(t) = \begin{cases} e^{t/2} & -1 \geq t > -5 \\ 0 & \text{otherwise} \end{cases}$$

and its time-reversed version $x(-t)$ is obtained by replacing t with $-t$ in $x(t)$ as

$$x(-t) = \begin{cases} e^{-t/2} & -1 \geq -t > -5 \quad \text{or} \quad 1 \leq t < 5 \\ 0 & \text{otherwise} \end{cases}$$

1.2-4 Combined Operations

Certain complex operations require simultaneous use of more than one of the operations just described. The most general operation involving all the three operations is $x(at - b)$, which is realized in two possible sequences of operation:

1. Time-shift $x(t)$ by b to obtain $x(t - b)$. Now time-scale the shifted signal $x(t - b)$ by a [i.e., replace t with at] to obtain $x(at - b)$.
2. Time-scale $x(t)$ by a to obtain $x(at)$. Now time-shift $x(at)$ by b/a [i.e., replace t with $t - (b/a)$] to obtain $x[a(t - b/a)] = x(at - b)$. In either case, if a is negative, time scaling involves time reversal.

For example, the signal $x(2t - 6)$ can be obtained in two ways. We can delay $x(t)$ by 6 to obtain $x(t - 6)$, and then time-compress this signal by factor 2 (replace t with $2t$) to obtain $x(2t - 6)$.

Alternately, we can first time-compress $x(t)$ by factor 2 to obtain $x(2t)$, then delay this signal by 3 (replace t with $t - 3$) to obtain $x(2t - 6)$.

1.3 CLASSIFICATION OF SIGNALS

Classification helps us better understand and utilize the items around us. Cars, for example, are classified as sports, offroad, family, and so forth. Knowing you have a sports car is useful in deciding whether to drive on a highway or on a dirt road. Knowing you want to drive up a mountain, you would probably choose an offroad vehicle over a family sedan. Similarly, there are several classes of signals. Some signal classes are more suitable for certain applications than others. Further, different signal classes often require different mathematical tools. Here we shall consider only the following classes of signals, which are suitable for the scope of this book:

1. Continuous-time and discrete-time signals
2. Analog and digital signals
3. Periodic and aperiodic signals
4. Energy and power signals
5. Deterministic and probabilistic signals

1.3-1 Continuous-Time and Discrete-Time Signals

A signal that is specified for a continuum of values of time t (Fig. 1.10a) is a *continuous-time signal*, and a signal that is specified only at discrete values of t (Fig. 1.10b) is a *discrete-time signal*. Telephone and video camera outputs are continuous-time signals, whereas the quarterly gross national product (GNP), monthly sales of a corporation, and stock market daily averages are discrete-time signals.

1.3-2 Analog and Digital Signals

The concept of continuous time is often confused with that of analog. The two are not the same. The same is true of the concepts of discrete time and digital. A signal whose amplitude can take on any value in a continuous range is an *analog signal*. This means that an analog signal amplitude can take on an infinite number of values. A *digital signal*, on the other hand, is one whose amplitude can take on only a finite number of values. Signals associated with a digital computer are digital because they take on only two values (binary signals). A digital signal whose amplitudes can take on M values is an M -ary signal of which binary ($M = 2$) is a special case. The terms *continuous time* and *discrete time* qualify the nature of a signal along the time (horizontal) axis. The terms *analog* and *digital*, on the other hand, qualify the nature of the signal amplitude (vertical axis). Figure 1.11 shows examples of signals of various types. It is clear that analog is not necessarily continuous-time and digital need not be discrete-time. Figure 1.11c shows an example of an analog discrete-time signal. An analog signal can be converted into a digital signal [analog-to-digital (A/D) conversion] through quantization (rounding off), as explained in Sec. 8.3.

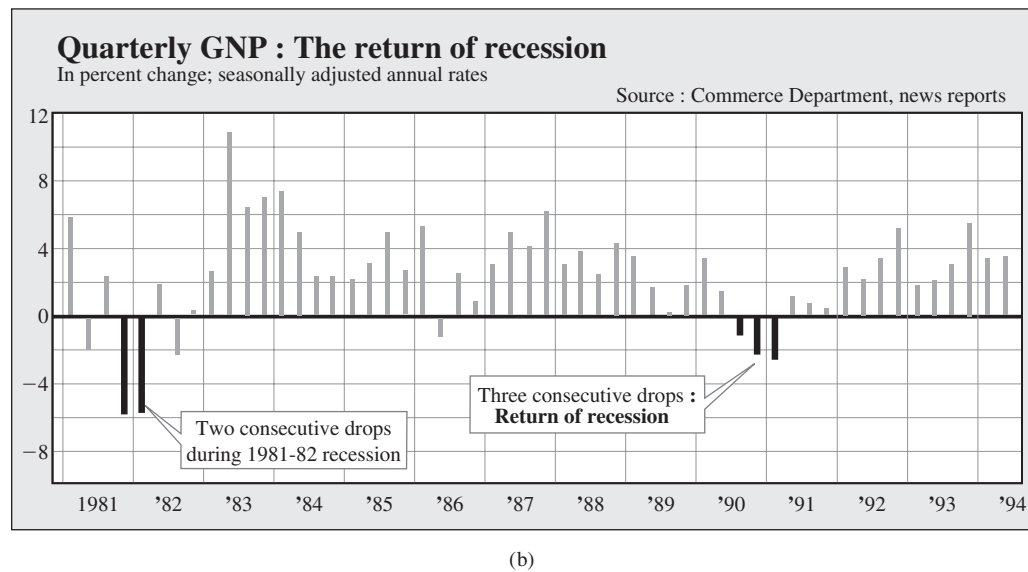
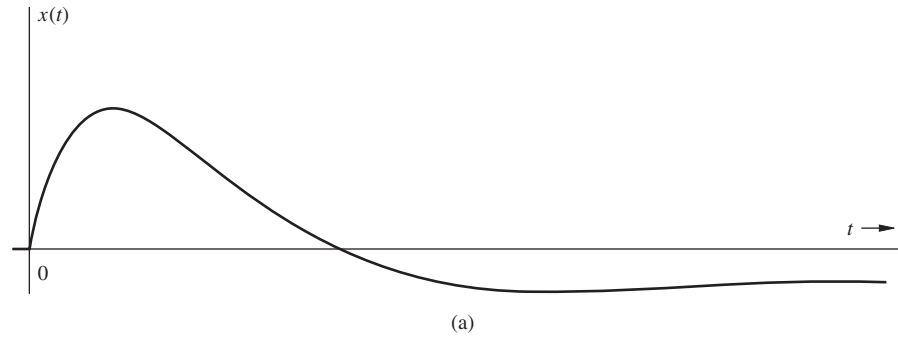


Figure 1.10 (a) Continuous-time and (b) discrete-time signals.

1.3-3 Periodic and Aperiodic Signals

A signal $x(t)$ is said to be *periodic* if for some positive constant T_0

$$x(t) = x(t + T_0) \quad \text{for all } t \quad (1.7)$$

The *smallest* value of T_0 that satisfies the periodicity condition of Eq. (1.7) is the *fundamental period* of $x(t)$. The signals in Figs. 1.2b and 1.3e are periodic signals with periods 2 and 1, respectively. A signal is *aperiodic* if it is not periodic. Signals in Figs. 1.2a, 1.3a, 1.3b, 1.3c, and 1.3d are all aperiodic.

By definition, a periodic signal $x(t)$ remains unchanged when time-shifted by one period. For this reason, a periodic signal must start at $t = -\infty$: if it started at some finite instant, say, $t = 0$, the time-shifted signal $x(t + T_0)$ would start at $t = -T_0$ and $x(t + T_0)$ would not be the same as

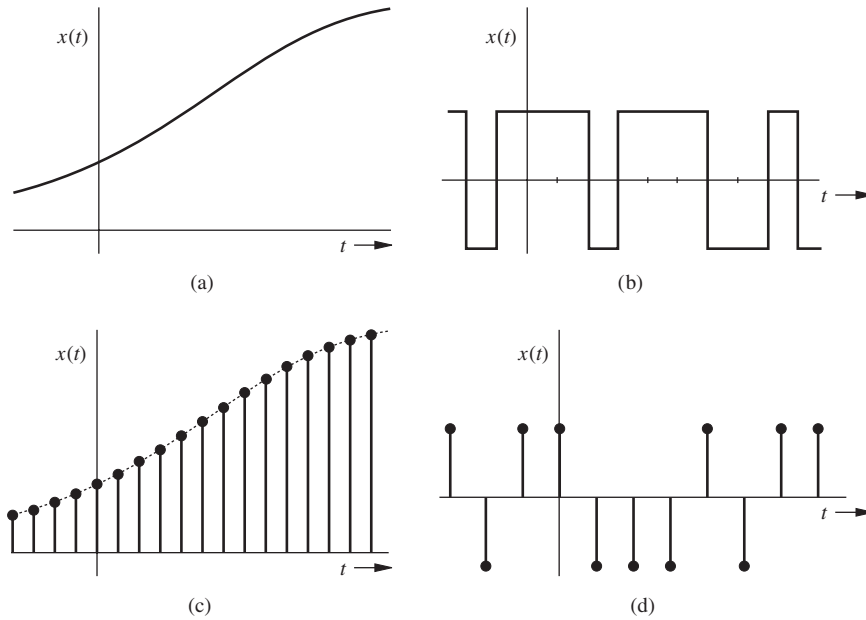


Figure 1.11 Examples of signals: **(a)** analog, continuous time; **(b)** digital, continuous time; **(c)** analog, discrete time; and **(d)** digital, discrete time.

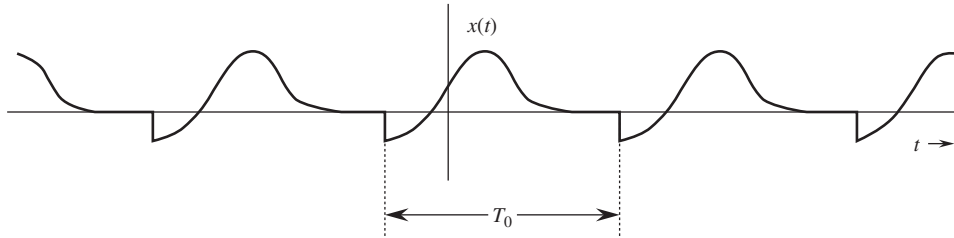


Figure 1.12 A periodic signal of period T_0 .

$x(t)$. Therefore, a *periodic signal*, by definition, must start at $t = -\infty$ and continue forever, as illustrated in Fig. 1.12.

Another important property of a periodic signal $x(t)$ is that $x(t)$ can be generated by *periodic extension* of any segment of $x(t)$ of duration T_0 (the period). As a result, we can generate $x(t)$ from any segment of $x(t)$ having a duration of one period by placing this segment and the reproduction thereof end to end ad infinitum on either side. Figure 1.13 shows a periodic signal $x(t)$ of period $T_0 = 6$. The shaded portion of Fig. 1.13a shows a segment of $x(t)$ starting at $t = -1$ and having a duration of one period (6 seconds). This segment, when repeated forever in either direction, results in the periodic signal $x(t)$. Figure 1.13b shows another shaded segment of $x(t)$ of duration T_0 starting at $t = 0$. Again, we see that this segment, when repeated forever on either side, results in $x(t)$. The reader can verify that this construction is possible with any segment of $x(t)$ starting at any instant as long as the segment duration is one period.

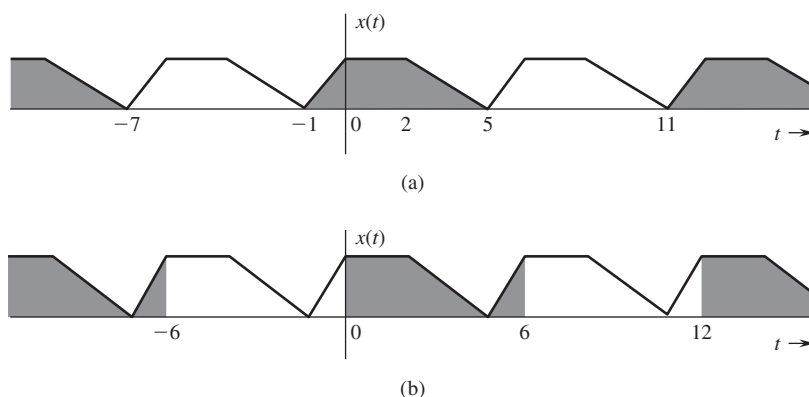


Figure 1.13 Generation of a periodic signal by periodic extension of its segment of one-period duration.

An additional useful property of a periodic signal $x(t)$ of period T_0 is that the area under $x(t)$ over any interval of duration T_0 is the same; that is, for any real numbers a and b ,

$$\int_a^{a+T_0} x(t) dt = \int_b^{b+T_0} x(t) dt$$

This result follows from the fact that a periodic signal takes the same values at the intervals of T_0 . Hence, the values over any segment of duration T_0 are repeated in any other interval of the same duration. For convenience, the area under $x(t)$ over any interval of duration T_0 will be denoted by

$$\int_{T_0} x(t) dt$$

It is helpful to label signals that start at $t = -\infty$ and continue forever as *everlasting* signals. Thus, an everlasting signal exists over the entire interval $-\infty < t < \infty$. The signals in Figs. 1.1b and 1.2b are examples of everlasting signals. Clearly, a periodic signal, by definition, is an everlasting signal.

A signal that does not start before $t = 0$ is a *causal* signal. In other words, $x(t)$ is a causal signal if

$$x(t) = 0 \quad t < 0$$

The signals in Figs. 1.3a–1.3c are causal signals. A signal that starts before $t = 0$ is a *noncausal* signal. All the signals in Figs. 1.1 and 1.2 are noncausal. Observe that an everlasting signal is always noncausal but a noncausal signal is not necessarily everlasting. The everlasting signal in Fig. 1.2b is noncausal; however, the noncausal signal in Fig. 1.2a is not everlasting. A signal that is zero for all $t \geq 0$ is called an *anti-causal* signal.

Comment. A true everlasting signal cannot be generated in practice for obvious reasons. Why should we bother to postulate such a signal? In later chapters we shall see that certain signals

(e.g., an impulse and an everlasting sinusoid) that cannot be generated in practice *do* serve a very useful purpose in the study of signals and systems.

1.3-4 Energy and Power Signals

A signal with finite energy is an *energy signal*, and a signal with finite and nonzero power is a *power signal*. The signals in Figs. 1.2a and 1.2b are examples of energy and power signals, respectively. Observe that power is the time average of energy. Since the averaging is over an infinitely large interval, a signal with finite energy has zero power, and a signal with finite power has infinite energy. Therefore, a signal cannot be both an energy signal and a power signal. If it is one, it cannot be the other. On the other hand, there are signals that are neither energy nor power signals. The ramp signal is one such case.

Comments. All practical signals have finite energies and are therefore energy signals. A power signal must necessarily have infinite duration; otherwise, its power, which is its energy averaged over an infinitely large interval, will not approach a (nonzero) limit. Clearly, it is impossible to generate a true power signal in practice because such a signal has infinite duration and infinite energy.

Also, because of periodic repetition, periodic signals for which the area under $|x(t)|^2$ over one period is finite are power signals; however, not all power signals are periodic.

DRILL 1.6 Neither Energy nor Power

Show that an everlasting exponential e^{-at} is neither an energy nor a power signal for any real value of a . However, if a is imaginary, it is a power signal with power $P_x = 1$ regardless of the value of a .

1.3-5 Deterministic and Random Signals

A signal whose physical description is known completely, in either a mathematical form or a graphical form, is a *deterministic signal*. A signal whose values cannot be predicted precisely but are known only in terms of probabilistic description, such as mean value or mean-squared value, is a *random signal*. In this book we shall exclusively deal with deterministic signals. Random signals are beyond the scope of this study.

1.4 SOME USEFUL SIGNAL MODELS

In the area of signals and systems, the step, the impulse, and the exponential functions play very important roles. Not only do they serve as a basis for representing other signals, but their use can simplify many aspects of the signals and systems.

1.4-1 The Unit Step Function $u(t)$

In much of our discussion, the signals begin at $t = 0$ (causal signals). Such signals can be conveniently described in terms of unit step function $u(t)$ shown in Fig. 1.14a. This function is defined by

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (1.8)$$

If we want a signal to start at $t = 0$ (so that it has a value of zero for $t < 0$), we need only multiply the signal by $u(t)$. For instance, the signal e^{-at} represents an everlasting exponential that starts at $t = -\infty$. The causal form of this exponential (Fig. 1.14b) can be described as $e^{-at}u(t)$.

The unit step function also proves very useful in specifying a function with different mathematical descriptions over different intervals. Examples of such functions appear in Fig. 1.7. These functions have different mathematical descriptions over different segments of time, as seen from Eqs. (1.5) and (1.6). Such a description often proves clumsy and inconvenient in mathematical treatment. We can use the unit step function to describe such functions by a single expression that is valid for all t .

Consider, for example, the rectangular pulse depicted in Fig. 1.15a. We can express such a pulse in terms of familiar step functions by observing that the pulse $x(t)$ can be expressed as the sum of the two delayed unit step functions, as shown in Fig. 1.15b. The unit step function $u(t)$ delayed by T seconds is $u(t - T)$. From Fig. 1.15b, it is clear that

$$x(t) = u(t - 2) - u(t - 4)$$

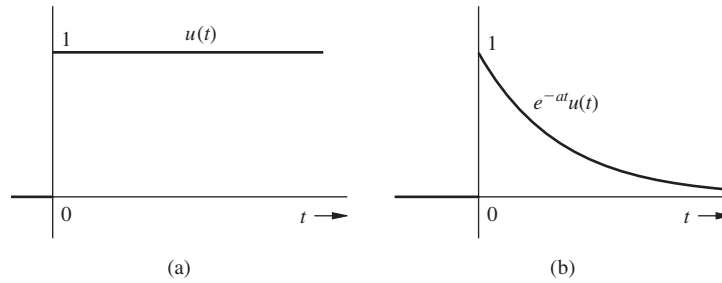


Figure 1.14 (a) Unit step function $u(t)$. (b) Exponential $e^{-at}u(t)$.

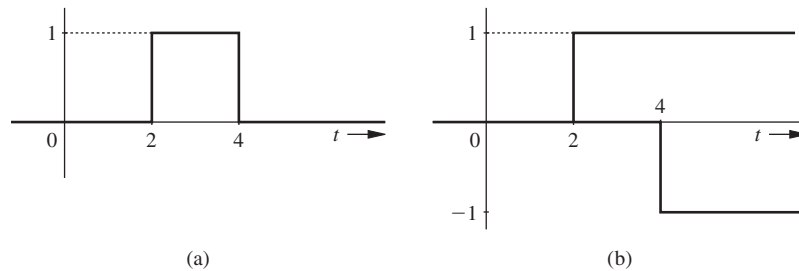


Figure 1.15 Representation of a rectangular pulse by step functions.

EXAMPLE 1.6 Describing a Triangle Function with the Unit Step

Use the unit step function to describe the signal in Fig. 1.16a.

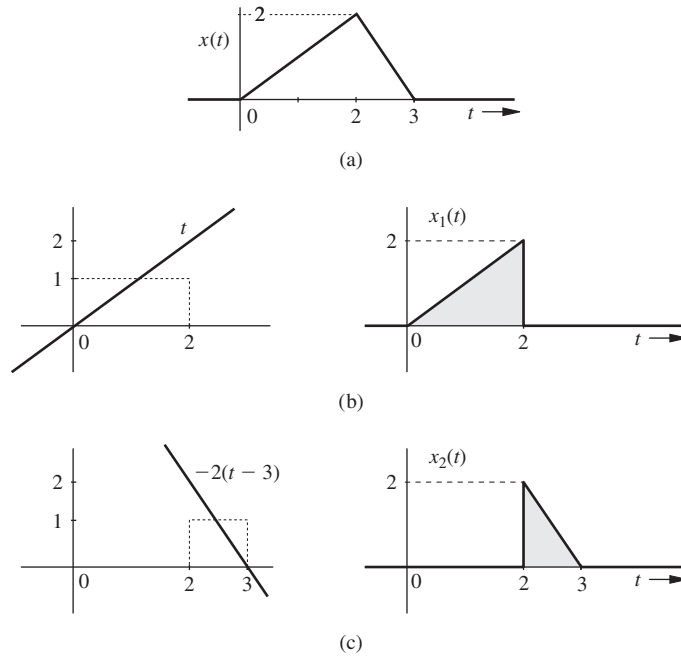


Figure 1.16 Representation of a signal defined interval by interval.

The signal illustrated in Fig. 1.16a can be conveniently handled by breaking it up into the two components $x_1(t)$ and $x_2(t)$, depicted in Figs. 1.16b and 1.16c, respectively. Here, $x_1(t)$ can be obtained by multiplying the ramp t by the gate pulse $u(t) - u(t-2)$, as shown in Fig. 1.16b. Therefore,

$$x_1(t) = t[u(t) - u(t-2)]$$

The signal $x_2(t)$ can be obtained by multiplying another ramp by the gate pulse illustrated in Fig. 1.16c. This ramp has a slope -2 ; hence it can be described by $-2t + c$. Now, because the ramp has a zero value at $t=3$, the constant $c=6$, and the ramp can be described by $-2(t-3)$. Also, the gate pulse in Fig. 1.16c is $u(t-2) - u(t-3)$. Therefore,

$$x_2(t) = -2(t-3)[u(t-2) - u(t-3)]$$

and

$$\begin{aligned}
 x(t) &= x_1(t) + x_2(t) \\
 &= t[u(t) - u(t-2)] - 2(t-3)[u(t-2) - u(t-3)] \\
 &= tu(t) - 3(t-2)u(t-2) + 2(t-3)u(t-3)
 \end{aligned}$$

EXAMPLE 1.7 Describing a Piecewise Function with the Unit Step

Describe the signal in Fig. 1.7a by a single expression valid for all t .

Over the interval from -1.5 to 0 , the signal can be described by a constant 2 , and over the interval from 0 to 3 , it can be described by $2e^{-t/2}$. Therefore,

$$\begin{aligned}
 x(t) &= \underbrace{2[u(t+1.5) - u(t)]}_{\text{constant part}} + \underbrace{2e^{-t/2}[u(t) - u(t-3)]}_{\text{exponential part}} \\
 &= 2u(t+1.5) - 2(1 - e^{-t/2})u(t) - 2e^{-t/2}u(t-3)
 \end{aligned}$$

Compare this expression with the expression for the same function found in Eq. (1.6).

DRILL 1.7 Using Reflected Unit Step Functions

Show that the signals depicted in Figs. 1.17a and 1.17b can be described as $u(-t)$ and $e^{-at}u(-t)$, respectively.

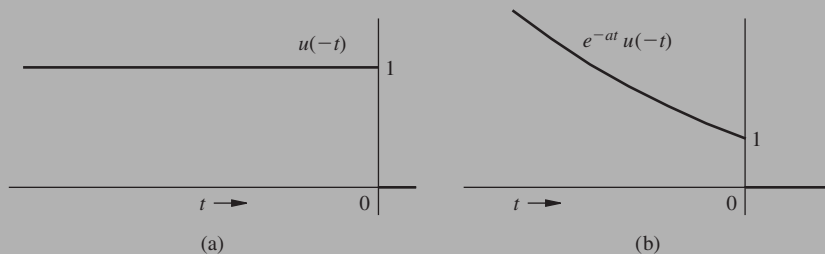


Figure 1.17 Signals for Drill 1.7.

DRILL 1.8 Describing a Piecewise Function with the Unit Step

Show that the signal shown in Fig. 1.18 can be described as

$$x(t) = (t - 1)u(t - 1) - (t - 2)u(t - 2) - u(t - 4)$$

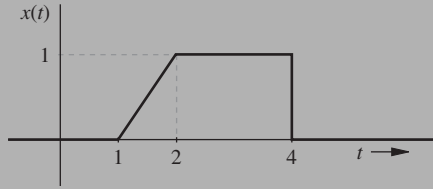


Figure 1.18 Signal for Drill 1.8.

1.4-2 The Unit Impulse Function $\delta(t)$

The unit impulse function $\delta(t)$ is one of the most important functions in the study of signals and systems. This function was first defined in two parts by P. A. M. Dirac as

$$\delta(t) = 0 \quad t \neq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (1.9)$$

We can visualize an impulse as a tall, narrow, rectangular pulse of unit area, as illustrated in Fig. 1.19b. The width of this rectangular pulse is a very small value $\epsilon \rightarrow 0$. Consequently, its height is a very large value $1/\epsilon \rightarrow \infty$. The unit impulse therefore can be regarded as a rectangular pulse with a width that has become infinitesimally small, a height that has become infinitely large, and an overall area that has been maintained at unity. Thus $\delta(t) = 0$ everywhere except at $t = 0$, where it is undefined. For this reason, a unit impulse is represented by the spearlike symbol in Fig. 1.19a.

Other pulses, such as the exponential, triangular, or Gaussian types, may also be used in impulse approximation. The important feature of the unit impulse function is not its shape but the fact that its effective duration (pulse width) approaches zero while its area remains at unity. For example, the exponential pulse $\alpha e^{-\alpha t} u(t)$ in Fig. 1.20a becomes taller and narrower as α increases.

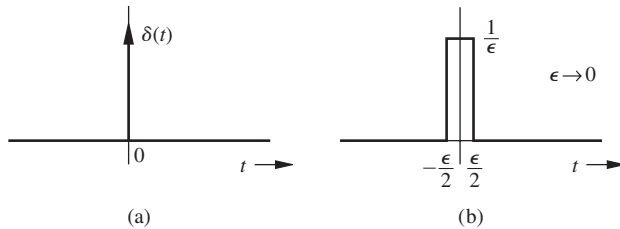


Figure 1.19 A unit impulse and its approximation.

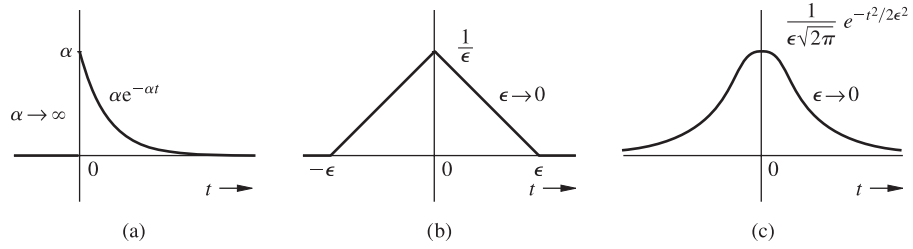


Figure 1.20 Other possible approximations to a unit impulse.

In the limit as $\alpha \rightarrow \infty$, the pulse height $\rightarrow \infty$, and its width or duration $\rightarrow 0$. Yet, the area under the pulse is unity regardless of the value of α because

$$\int_0^{\infty} \alpha e^{-\alpha t} dt = 1$$

The pulses in Figs. 1.20b and 1.20c behave in a similar fashion. Clearly, the exact impulse function cannot be generated in practice; it can only be approached.

From Eq. (1.9), it follows that the function $k\delta(t) = 0$ for all $t \neq 0$, and its area is k . Thus, $k\delta(t)$ is an impulse function whose area is k (in contrast to the unit impulse function, whose area is 1).

MULTIPLICATION OF A FUNCTION BY AN IMPULSE

Let us now consider what happens when we multiply the unit impulse $\delta(t)$ by a function $\phi(t)$ that is known to be continuous at $t = 0$. Since the impulse has nonzero value only at $t = 0$, and the value of $\phi(t)$ at $t = 0$ is $\phi(0)$, we obtain

$$\phi(t)\delta(t) = \phi(0)\delta(t)$$

Thus, multiplication of a continuous-time function $\phi(t)$ with an unit impulse located at $t = 0$ results in an impulse, which is located at $t = 0$ and has strength $\phi(0)$ [the value of $\phi(t)$ at the location of the impulse]. Use of exactly the same argument leads to the generalization of this result, stating that provided $\phi(t)$ is continuous at $t = T$, $\phi(t)$ multiplied by an impulse $\delta(t - T)$ (impulse located at $t = T$) results in an impulse located at $t = T$ and having strength $\phi(T)$ [the value of $\phi(t)$ at the location of the impulse].

$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T) \quad (1.10)$$

SAMPLING PROPERTY OF THE UNIT IMPULSE FUNCTION

From Eq. (1.10) it follows that

$$\int_{-\infty}^{\infty} \phi(t)\delta(t - T) dt = \phi(T) \int_{-\infty}^{\infty} \delta(t) dt = \phi(T) \quad (1.11)$$

provided $\phi(t)$ is continuous at $t = T$. This result means that *the area under the product of a function with an impulse $\delta(t - T)$ is equal to the value of that function at the instant at which the unit impulse is located*. This property is very important and useful and is known as the *sampling* or *sifting* property of the unit impulse.

UNIT IMPULSE AS A GENERALIZED FUNCTION

The definition of the unit impulse function given in Eq. (1.9) is not mathematically rigorous, which leads to serious difficulties. First, the impulse function does not define a unique function: for example, it can be shown that $\delta(t) + \dot{\delta}(t)$ also satisfies Eq. (1.9) [1]. Moreover, $\delta(t)$ is not even a true function in the ordinary sense. An ordinary function is specified by its values for all time t . The impulse function is zero everywhere except at $t = 0$, and at this, the only interesting part of its range, it is undefined. These difficulties are resolved by defining the impulse as a generalized function rather than an ordinary function. A *generalized function* is defined by its effect on other functions instead of by its value at every instant of time.

In this approach the impulse function is defined by the sampling property [Eq. (1.11)]. We say nothing about what the impulse function is or what it looks like. Instead, the impulse function is defined in terms of its effect on a test function $\phi(t)$. We define a unit impulse as a function for which the area under its product with a function $\phi(t)$ is equal to the value of the function $\phi(t)$ at the instant at which the impulse is located. It is assumed that $\phi(t)$ is continuous at the location of the impulse. Recall that the sampling property [Eq. (1.11)] is the consequence of the classical (Dirac) definition of the unit impulse in Eq. (1.9). In contrast, *the sampling property [Eq. (1.11)] defines the impulse function in the generalized function approach.*

We now present an interesting application of the generalized function definition of an impulse. Because the unit step function $u(t)$ is discontinuous at $t = 0$, its derivative du/dt does not exist at $t = 0$ in the ordinary sense. We now show that this derivative *does* exist in the generalized sense, and it is, in fact, $\delta(t)$. As a proof, let us evaluate the integral of $(du/dt)\phi(t)$, using integration by parts:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{du(t)}{dt} \phi(t) dt &= u(t)\phi(t) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} u(t)\dot{\phi}(t) dt \\ &= \phi(\infty) - 0 - \int_0^{\infty} \dot{\phi}(t) dt \\ &= \phi(\infty) - \phi(t) \Big|_0^{\infty} = \phi(0) \end{aligned}$$

This result shows that du/dt satisfies the sampling property of $\delta(t)$. Therefore it is an impulse $\delta(t)$ in the generalized sense—that is,

$$\frac{du(t)}{dt} = \delta(t) \quad (1.12)$$

Consequently,

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

These results can also be obtained graphically from Fig. 1.19b. We observe that the area from $-\infty$ to t under the limiting form of $\delta(t)$ in Fig. 1.19b is zero if $t < -\epsilon/2$ and unity if $t \geq \epsilon/2$ with $\epsilon \rightarrow 0$. Consequently,

$$\begin{aligned} \int_{-\infty}^t \delta(\tau) d\tau &= \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \\ &= u(t) \end{aligned}$$

This result shows that the unit step function can be obtained by integrating the unit impulse function. Similarly the unit ramp function $x(t) = tu(t)$ can be obtained by integrating the unit step function. We may continue with unit parabolic function $t^2/2$ obtained by integrating the unit ramp, and so on. On the other side, we have derivatives of impulse function, which can be defined as generalized functions (see Prob. 1.4-12). All these functions, derived from the unit impulse function (successive derivatives and integrals), are called *singularity functions*.[†]

DRILL 1.9 Simplifying Expressions Containing the Unit Impulse

Show that

- (a) $(t^3 + 3)\delta(t) = 3\delta(t)$
- (b) $\left[\sin\left(t^2 - \frac{\pi}{2}\right)\right]\delta(t) = -\delta(t)$
- (c) $e^{-2t}\delta(t) = \delta(t)$
- (d) $\frac{\omega^2 + 1}{\omega^2 + 9}\delta(\omega - 1) = \frac{1}{5}\delta(\omega - 1)$

DRILL 1.10 Simplifying Integrals Containing the Unit Impulse

Show that

- (a) $\int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = 1$
- (b) $\int_{-\infty}^{\infty} \delta(t-2)\cos\left(\frac{\pi t}{4}\right) dt = 0$
- (c) $\int_{-\infty}^{\infty} e^{-2(x-t)}\delta(2-t) dt = e^{-2(x-2)}$

1.4-3 The Exponential Function e^{st}

Another important function in the area of signals and systems is the exponential signal e^{st} , where s is complex in general, given by

$$s = \sigma + j\omega$$

[†] Singularity functions were defined by late Prof. S. J. Mason as follows. A singularity is a point at which a function does not possess a derivative. Each of the singularity functions (or if not the function itself, then the function differentiated a finite number of times) has a singular point at the origin and is zero elsewhere [2].

Therefore,

$$e^{st} = e^{(\sigma + j\omega)t} = e^{\sigma t} e^{j\omega t} = e^{\sigma t} (\cos \omega t + j \sin \omega t) \quad (1.13)$$

Since $s^* = \sigma - j\omega$ (the conjugate of s), then

$$e^{s^*t} = e^{(\sigma - j\omega)t} = e^{\sigma t} e^{-j\omega t} = e^{\sigma t} (\cos \omega t - j \sin \omega t)$$

and

$$e^{\sigma t} \cos \omega t = \frac{1}{2}(e^{st} + e^{s^*t}) \quad (1.14)$$

A comparison of Eq. (1.13) with Euler's formula shows that e^{st} is a generalization of the function $e^{j\omega t}$, where the frequency variable $j\omega$ is generalized to a complex variable $s = \sigma + j\omega$. For this reason, we designate the variable s as the *complex frequency*. In fact, function e^{st} encompasses a large class of functions. The following functions are either special cases of or can be expressed in terms of e^{st} :

1. A constant $k = ke^{0t}$ ($s = 0$)
2. A monotonic exponential $e^{\sigma t}$ ($\omega = 0, s = \sigma$)
3. A sinusoid $\cos \omega t$ ($\sigma = 0, s = \pm j\omega$)
4. An exponentially varying sinusoid $e^{\sigma t} \cos \omega t$ ($s = \sigma \pm j\omega$)

These functions are illustrated in Fig. 1.21.

The complex frequency s can be conveniently represented on a *complex frequency plane* (s plane), as depicted in Fig. 1.22. The horizontal axis is the real axis (σ axis), and the vertical axis is the imaginary axis (ω axis). The absolute value of the imaginary part of s is $|\omega|$ (the

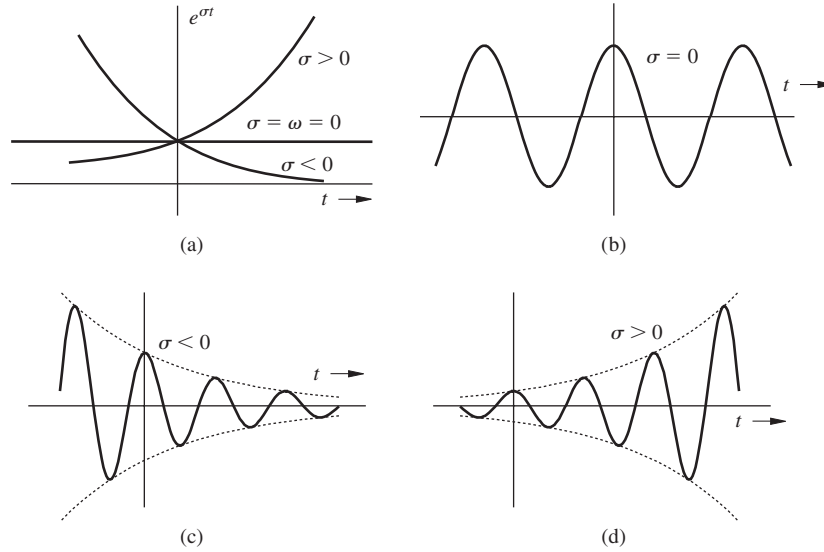


Figure 1.21 Sinusoids of complex frequency $\sigma + j\omega$.

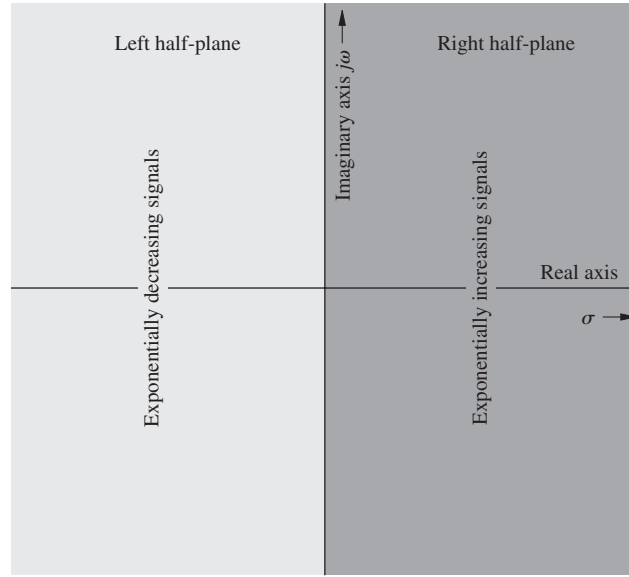


Figure 1.22 Complex frequency plane.

radian frequency), which indicates the frequency of oscillation of e^{st} ; the real part σ (the *neper* frequency) gives information about the rate of increase or decrease of the amplitude of e^{st} . For signals whose complex frequencies lie on the real axis (σ axis, where $\omega = 0$), the frequency of oscillation is zero. Consequently these signals are monotonically increasing or decreasing exponentials (Fig. 1.21a). For signals whose frequencies lie on the imaginary axis (ω axis, where $\sigma = 0$), $e^{\sigma t} = 1$. Therefore, these signals are conventional sinusoids with constant amplitude (Fig. 1.21b). The case $s = 0$ ($\sigma = \omega = 0$) corresponds to a constant (dc) signal because $e^{0t} = 1$. For the signals illustrated in Figs. 1.21c and 1.21d, both σ and ω are nonzero; the frequency s is complex and does not lie on either axis. The signal in Fig. 1.21c decays exponentially. Therefore, σ is negative, and s lies to the left of the imaginary axis. In contrast, the signal in Fig. 1.21d *grows* exponentially. Therefore, σ is positive, and s lies to the right of the imaginary axis. Thus the s plane (Fig. 1.21) can be separated into two parts: the *left half-plane* (LHP) corresponding to exponentially decaying signals and the *right half-plane* (RHP) corresponding to exponentially growing signals. The imaginary axis separates the two regions and corresponds to signals of constant amplitude.

An exponentially growing sinusoid $e^{2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(2+j5)t}$ and $e^{(2-j5)t}$ with complex frequencies $2 + j5$ and $2 - j5$, respectively, which lie in the RHP. An exponentially decaying sinusoid $e^{-2t} \cos 5t$ can be expressed as a linear combination of exponentials $e^{(-2+j5)t}$ and $e^{(-2-j5)t}$ with complex frequencies $-2 + j5$ and $-2 - j5$, respectively, which lie in the LHP. A constant-amplitude sinusoid $\cos 5t$ can be expressed as a linear combination of exponentials e^{j5t} and e^{-j5t} with complex frequencies $\pm j5$, which lie on the imaginary axis. Observe that the monotonic exponentials $e^{\pm 2t}$ are also generalized sinusoids with complex frequencies ± 2 .

1.5 EVEN AND ODD FUNCTIONS

A function $x_e(t)$ is said to be an *even function* of t if it is symmetrical about the vertical axis. A function $x_o(t)$ is said to be an *odd function* of t if it is antisymmetrical about the vertical axis. Mathematically expressed, these symmetry conditions require

$$x_e(t) = x_e(-t) \quad \text{and} \quad x_o(t) = -x_o(-t) \quad (1.15)$$

An even function has the same value at the instants t and $-t$ for all values of t . On the other hand, the value of an odd function at the instant t is the negative of its value at the instant $-t$. An example even signal and an example odd signal are shown in Figs. 1.23a and 1.23b, respectively.

1.5-1 Some Properties of Even and Odd Functions

Even and odd functions have the following properties:

even function $\times$ odd function = odd function

odd function $\times$ odd function = even function

even function $\times$ even function = even function

The proofs are trivial and follow directly from the definition of odd and even functions [Eq. (1.15)].

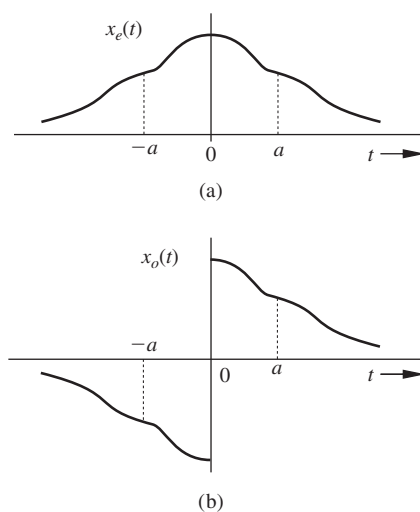


Figure 1.23 Functions of t : (a) even and (b) odd.

AREA

Because of the symmetries of even and odd functions about the vertical axis, it follows from Eq. (1.15) [or Fig. 1.23] that

$$\int_{-a}^a x_e(t) dt = 2 \int_0^a x_e(t) dt \quad \text{and} \quad \int_{-a}^a x_o(t) dt = 0 \quad (1.16)$$

These results are valid under the assumption that there is no impulse (or its derivatives) at the origin. The proof of these statements is obvious from the plots of even and odd functions. Formal proofs, left as an exercise for the reader, can be accomplished by using the definitions in Eq. (1.15).

Because of their properties, study of odd and even functions proves useful in many applications, as will become evident in later chapters.

1.5-2 Even and Odd Components of a Signal

Every signal $x(t)$ can be expressed as a sum of even and odd components because

$$x(t) = \underbrace{\frac{1}{2}[x(t) + x(-t)]}_{\text{even}} + \underbrace{\frac{1}{2}[x(t) - x(-t)]}_{\text{odd}} \quad (1.17)$$

From the definitions in Eq. (1.15), we can clearly see that the first component on the right-hand side is an even function, while the second component is odd. This is apparent from the fact that replacing t by $-t$ in the first component yields the same function. The same maneuver in the second component yields the negative of that component.

EXAMPLE 1.8 Finding the Even and Odd Components of a Signal

Find and sketch the even and odd components of $x(t) = e^{-at}u(t)$.

Based on Eq. (1.17), we can express $x(t)$ as a sum of the even component $x_e(t)$ and the odd component $x_o(t)$ as

$$x(t) = x_e(t) + x_o(t)$$

where

$$x_e(t) = \frac{1}{2}[e^{-at}u(t) + e^{at}u(-t)] \quad \text{and} \quad x_o(t) = \frac{1}{2}[e^{-at}u(t) - e^{at}u(-t)]$$

The function $e^{-at}u(t)$ and its even and odd components are illustrated in Fig. 1.24.

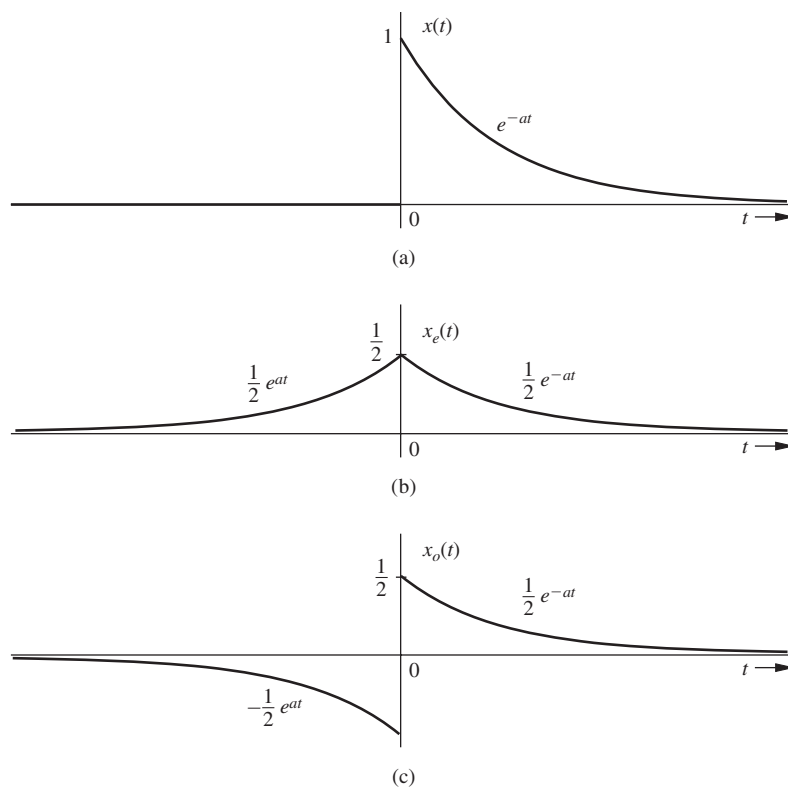


Figure 1.24 Finding even and odd components of a signal.

EXAMPLE 1.9 Finding the Even and Odd Components of a Complex Signal

Find the even and odd components of e^{jt} .

From Eq. (1.17),

$$e^{jt} = x_e(t) + x_o(t)$$

where

$$x_e(t) = \frac{1}{2}[e^{jt} + e^{-jt}] = \cos t \quad \text{and} \quad x_o(t) = \frac{1}{2}[e^{jt} - e^{-jt}] = j \sin t$$

A MODIFICATION FOR COMPLEX SIGNALS

While a complex signal can be decomposed into even and odd components, it is more common to decompose complex signals using conjugate symmetries. A complex signal $x(t)$ is said to be *conjugate-symmetric* if $x(t) = x^*(-t)$. A conjugate-symmetric signal is even in the real part and odd in the imaginary part. Thus, a real conjugate-symmetric signal is an even signal. A signal is *conjugate-antisymmetric* if $x(t) = -x^*(-t)$. A conjugate-antisymmetric signal is odd in the real part and even in the imaginary part. A real conjugate-antisymmetric signal is an odd signal. Any signal $x(t)$ can be decomposed into a conjugate-symmetric portion $x_{cs}(t)$ plus a conjugate-antisymmetric portion $x_{ca}(t)$. That is,

$$x(t) = x_{cs}(t) + x_{ca}(t)$$

where

$$x_{cs}(t) = \frac{x(t) + x^*(-t)}{2} \quad \text{and} \quad x_{ca}(t) = \frac{x(t) - x^*(-t)}{2}$$

The proof is similar to the one for decomposing a signal into even and odd components. As we shall see in later chapters, conjugate symmetries commonly occur in real-world signals and their transforms.

1.6 SYSTEMS

As mentioned in Sec. 1.1, systems are used to process signals to allow modification or extraction of additional information from the signals. A system may consist of physical components (hardware realization) or of an algorithm that computes the output signal from the input signal (software realization).

Roughly speaking, a physical system consists of interconnected components, which are characterized by their terminal (input–output) relationships. In addition, a system is governed by laws of interconnection. For example, in electrical systems, the terminal relationships are the familiar voltage–current relationships for the resistors, capacitors, inductors, transformers, transistors, and so on, as well as the laws of interconnection (i.e., Kirchhoff’s laws). We use these laws to derive mathematical equations relating the outputs to the inputs. These equations then represent a *mathematical model* of the system.

A system can be conveniently illustrated by a “black box” with one set of accessible terminals where the input variables $x_1(t), x_2(t), \dots, x_j(t)$ are applied and another set of accessible terminals where the output variables $y_1(t), y_2(t), \dots, y_k(t)$ are observed (Fig. 1.25).

The study of systems consists of three major areas: mathematical modeling, analysis, and design. Although we shall be dealing with mathematical modeling, our main concern is with

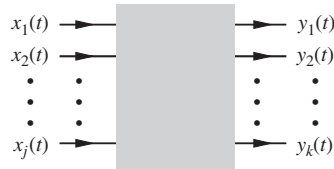


Figure 1.25 Representation of a system.

analysis and design. The major portion of this book is devoted to the analysis problem—how to determine the system outputs for the given inputs and a given mathematical model of the system (or rules governing the system). To a lesser extent, we will also consider the problem of design or synthesis—how to construct a system that will produce a desired set of outputs for the given inputs.

DATA NEEDED TO COMPUTE SYSTEM RESPONSE

To understand what data we need to compute a system response, consider a simple RC circuit with a current source $x(t)$ as its input (Fig. 1.26).

The output voltage $y(t)$ is given by

$$y(t) = Rx(t) + \frac{1}{C} \int_{-\infty}^t x(\tau) d\tau \quad (1.18)$$

The limits of the integral on the right-hand side are from $-\infty$ to t because this integral represents the capacitor charge due to the current $x(t)$ flowing in the capacitor, and this charge is the result of the current flowing in the capacitor from $-\infty$. Now, Eq. (1.18) can be expressed as

$$y(t) = Rx(t) + \frac{1}{C} \int_{-\infty}^0 x(\tau) d\tau + \frac{1}{C} \int_0^t x(\tau) d\tau$$

The middle term on the right-hand side is $v_C(0)$, the capacitor voltage at $t = 0$. Therefore,

$$y(t) = v_C(0) + Rx(t) + \frac{1}{C} \int_0^t x(\tau) d\tau \quad t \geq 0$$

This equation can be readily generalized as

$$y(t) = v_C(t_0) + Rx(t) + \frac{1}{C} \int_{t_0}^t x(\tau) d\tau \quad t \geq t_0 \quad (1.19)$$

From Eq. (1.18), the output voltage $y(t)$ at an instant t can be computed if we know the input current flowing in the capacitor throughout its entire past ($-\infty$ to t). Alternatively, if we know the input current $x(t)$ from some moment t_0 onward, then, using Eq. (1.19), we can still calculate $y(t)$ for $t \geq t_0$ from a knowledge of the input current, provided we know $v_C(t_0)$, the initial capacitor voltage (voltage at t_0). Thus $v_C(t_0)$ contains all the relevant information about the circuit's entire

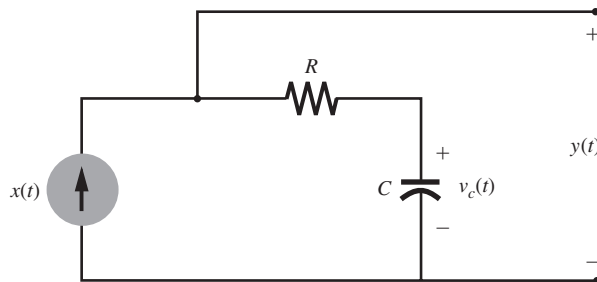


Figure 1.26 Example of a simple electrical system.

past ($-\infty$ to t_0) that we need to compute $y(t)$ for $t \geq t_0$. Therefore, the response of a system at $t \geq t_0$ can be determined from its input(s) during the interval t_0 to t and from certain *initial conditions* at $t = t_0$.

In the preceding example, we needed only one initial condition. However, in more complex systems, several initial conditions may be necessary. We know, for example, that in passive *RLC* networks, the initial values of all inductor currents and all capacitor voltages[†] are needed to determine the outputs at any instant $t \geq 0$ if the inputs are given over the interval $[0, t]$.

1.7 CLASSIFICATION OF SYSTEMS

Systems may be classified broadly in the following categories:

1. Linear and nonlinear systems
2. Constant-parameter and time-varying-parameter systems
3. Instantaneous (memoryless) and dynamic (with memory) systems
4. Causal and noncausal systems
5. Continuous-time and discrete-time systems
6. Analog and digital systems
7. Invertible and noninvertible systems
8. Stable and unstable systems

Other classifications, such as deterministic and probabilistic systems, are beyond the scope of this text and are not considered.

1.7-1 Linear and Nonlinear Systems

THE CONCEPT OF LINEARITY

A system whose output is proportional to its input is an *example* of a linear system. But linearity implies more than this; it also implies the *additivity property*: that is, if several inputs are acting on a system, then the total effect on the system due to all these inputs can be determined by considering one input at a time while assuming all the other inputs to be zero. The total effect is then the sum of all the component effects. This property may be expressed as follows: for a linear system, if an input x_1 acting alone has an effect y_1 , and if another input x_2 , also acting alone, has an effect y_2 , then, with both inputs acting on the system, the total effect will be $y_1 + y_2$. Thus, if

$$x_1 \longrightarrow y_1 \quad \text{and} \quad x_2 \longrightarrow y_2$$

then for all x_1 and x_2

$$x_1 + x_2 \longrightarrow y_1 + y_2 \tag{1.20}$$

In addition, a linear system must satisfy the *homogeneity* or scaling property, which states that for arbitrary real or imaginary number k , if an input is increased k -fold, the effect also increases k -fold. Thus, if

$$x \longrightarrow y$$

[†] Strictly speaking, this means independent inductor currents and capacitor voltages.

then for all real or imaginary k

$$kx \longrightarrow ky \quad (1.21)$$

Thus, linearity implies two properties: homogeneity (scaling) and additivity.[†] Both these properties can be combined into one property (*superposition*), which is expressed as follows: If

$$x_1 \longrightarrow y_1 \quad \text{and} \quad x_2 \longrightarrow y_2$$

then for all inputs x_1 and x_2 and all constants k_1 and k_2 ,

$$k_1x_1 + k_2x_2 \longrightarrow k_1y_1 + k_2y_2 \quad (1.22)$$

There is another useful way to view the linearity condition described in Eq. (1.22): the response of a linear system is unchanged whether the operations of summing and scaling precede the system (sum and scale act on inputs) or follow the system (sum and scale act on outputs). *Thus, linearity implies commutability between a system and the operations of summing and scaling.* It may appear that additivity implies homogeneity. Unfortunately, homogeneity does not always follow from additivity. Drill 1.11 demonstrates such a case.

DRILL 1.11 Additivity but Not Homogeneity

Show that a system with the input $x(t)$ and the output $y(t)$ related by $y(t) = \text{Re}\{x(t)\}$ satisfies the additivity property but violates the homogeneity property. Hence, such a system is not linear. [Hint: Show that Eq. (1.21) is not satisfied when k is complex.]

RESPONSE OF A LINEAR SYSTEM

For the sake of simplicity, we discuss only *single-input, single-output (SISO)* systems. But the discussion can be readily extended to *multiple-input, multiple-output (MIMO)* systems.

A system's output for $t \geq 0$ is the result of two independent causes: the initial conditions of the system (or the system state) at $t = 0$ and the input $x(t)$ for $t \geq 0$. If a system is to be linear, the output must be the sum of the two components resulting from these two causes: first, the *zero-input response (ZIR)* that results only from the initial conditions at $t = 0$ with the input $x(t) = 0$ for $t \geq 0$, and then the *zero-state response (ZSR)* that results only from the input $x(t)$ for $t \geq 0$ when the initial conditions (at $t = 0$) are assumed to be zero. When all the appropriate initial conditions are zero, the system is said to be in *zero state*. The system output is zero when the input is zero only if the system is in zero state.

In summary, a linear system response can be expressed as the sum of the zero-input and zero-state responses:

$$\text{total response} = \text{zero-input response} + \text{zero-state response}$$

[†] A linear system must also satisfy the additional condition of *smoothness*, where small changes in the system's inputs must result in small changes in its outputs [3].

This property of linear systems, which permits the separation of an output into components resulting from the initial conditions and from the input, is called the *decomposition property*. For the RC circuit of Fig. 1.26, the response $y(t)$ was found to be [see Eq. (1.19) with $t_0 = 0$]

$$y(t) = \underbrace{v_C(0)}_{\text{ZIR}} + \underbrace{Rx(t) + \frac{1}{C} \int_0^t x(\tau) d\tau}_{\text{ZSR}} \quad (1.23)$$

From Eq. (1.23), it is clear that if the input $x(t) = 0$ for $t \geq 0$, the output $y(t) = v_C(0)$. Hence $v_C(0)$ is the zero-input response of the response $y(t)$. Similarly, if the system state (the voltage v_C in this case) is zero at $t = 0$, the output is given by the second component on the right-hand side of Eq. (1.23). Clearly this is the zero-state response of the response $y(t)$.

In addition to the decomposition property, linearity implies that both the zero-input and zero-state components must obey the principle of superposition with respect to each of their respective causes. For example, if we increase the initial condition k -fold, the zero-input response must also increase k -fold. Similarly, if we increase the input k -fold, the zero-state response must also increase k -fold. These facts can be readily verified from Eq. (1.23) for the RC circuit in Fig. 1.26. For instance, if we double the initial condition $v_C(0)$, the zero-input response doubles; if we double the input $x(t)$, the zero-state response doubles.

EXAMPLE 1.10 Linearity of Constant-Coefficient Linear Differential Equations

Show that the system described by the equation

$$\frac{dy(t)}{dt} + 3y(t) = x(t) \quad (1.24)$$

is linear.

Let the system response to the inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. Then

$$\frac{dy_1(t)}{dt} + 3y_1(t) = x_1(t) \quad \text{and} \quad \frac{dy_2(t)}{dt} + 3y_2(t) = x_2(t)$$

Multiplying the first equation by k_1 , the second by k_2 , and adding them yield

$$\frac{d}{dt}[k_1y_1(t) + k_2y_2(t)] + 3[k_1y_1(t) + k_2y_2(t)] = k_1x_1(t) + k_2x_2(t)$$

But this equation is the system equation [Eq. (1.24)] with

$$x(t) = k_1x_1(t) + k_2x_2(t) \quad \text{and} \quad y(t) = k_1y_1(t) + k_2y_2(t)$$

Therefore, when the input is $k_1x_1(t) + k_2x_2(t)$, the system response is $k_1y_1(t) + k_2y_2(t)$. Consequently, the system is linear. Using this argument, we can readily generalize the result to

show that a system described by a differential equation of the form

$$a_0 \frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \cdots + a_N y(t) = b_{N-M} \frac{d^M x(t)}{dt^M} + \cdots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t) \quad (1.25)$$

is a linear system. The coefficients a_i and b_i in this equation can be constants or functions of time. Although here we proved only zero-state linearity, it can be shown that such systems are also zero-input linear and have the decomposition property.

DRILL 1.12 Linearity of a Differential Equation with Time-Varying Parameters

Show that the system described by the following equation is linear:

$$\frac{dy(t)}{dt} + t^2 y(t) = (2t + 3)x(t)$$

DRILL 1.13 A Nonlinear Differential Equation

Show that the system described by the following equation is nonlinear:

$$y(t) \frac{dy(t)}{dt} + 3y(t) = x(t)$$

MORE COMMENTS ON LINEAR SYSTEMS

Almost all systems observed in practice become nonlinear when large enough signals are applied to them. However, it is possible to approximate most of the nonlinear systems by linear systems for small-signal analysis. The analysis of nonlinear systems is generally difficult. Nonlinearities can arise in so many ways that describing them with a common mathematical form is impossible. Not only is each system a category in itself, but even for a given system, changes in initial conditions or input amplitudes may change the nature of the problem. On the other hand, the superposition property of linear systems is a powerful unifying principle that allows for a general solution. The superposition property (linearity) greatly simplifies the analysis of linear systems. Because of the decomposition property, we can evaluate separately the two components of the output. The zero-input response can be computed by assuming the input to be zero, and the zero-state response can be computed by assuming zero initial conditions. Moreover, if we express an

input $x(t)$ as a sum of simpler functions,

$$x(t) = a_1x_1(t) + a_2x_2(t) + \cdots + a_mx_m(t)$$

then, by virtue of linearity, the response $y(t)$ is given by

$$y(t) = a_1y_1(t) + a_2y_2(t) + \cdots + a_my_m(t)$$

where $y_k(t)$ is the zero-state response to an input $x_k(t)$. This apparently trivial observation has profound implications. As we shall see repeatedly in later chapters, it proves extremely useful and opens new avenues for analyzing linear systems.

For example, consider an arbitrary input $x(t)$ such as the one shown in Fig. 1.27a. We can approximate $x(t)$ with a sum of rectangular pulses of width Δt and of varying heights. The approximation improves as $\Delta t \rightarrow 0$, when the rectangular pulses become impulses spaced Δt seconds apart (with $\Delta t \rightarrow 0$).[†] Thus, an arbitrary input can be replaced by a weighted sum of impulses spaced Δt ($\Delta t \rightarrow 0$) seconds apart. Therefore, if we know the system response to a unit impulse, we can immediately determine the system response to an arbitrary input $x(t)$ by adding the system response to each impulse component of $x(t)$. A similar situation is depicted in Fig. 1.27b, where $x(t)$ is approximated by a sum of step functions of varying magnitude and spaced Δt seconds apart. The approximation improves as Δt becomes smaller. Therefore, if we know the system response to a unit step input, we can compute the system response to any arbitrary input $x(t)$ with relative ease. Time-domain analysis of linear systems (discussed in Ch. 2) uses this approach.

Chapters 4, 5, 6, and 7 employ the same approach but instead use sinusoids or exponentials as the basic signal components. We show that any arbitrary input signal can be expressed as a weighted sum of sinusoids (or exponentials) having various frequencies. Thus a knowledge of the system response to a sinusoid enables us to determine the system response to an arbitrary input $x(t)$.

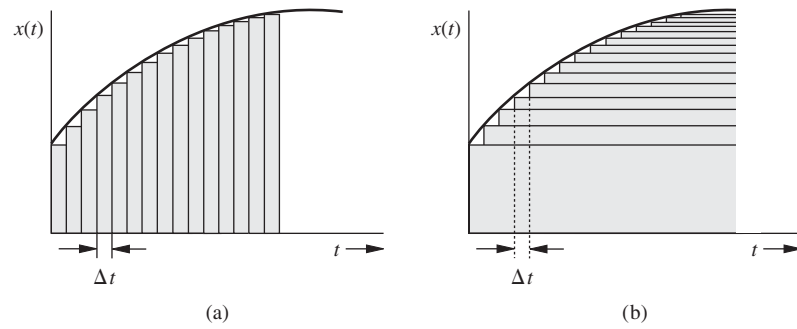


Figure 1.27 Signal representation in terms of impulse and step components.

[†] Here, the discussion of a rectangular pulse approaching an impulse at $\Delta t \rightarrow 0$ is somewhat imprecise. It is explained in Sec. 2.4 with more rigor.

1.7-2 Time-Invariant and Time-Varying Systems

Systems whose parameters do not change with time are *time-invariant* (also *constant-parameter*) systems. For such a system, if the input is delayed by T seconds, the output is the same as before but delayed by T (assuming initial conditions are also delayed by T). This property is expressed graphically in Fig. 1.28. We can also illustrate this property, as shown in Fig. 1.29. We can delay the output $y(t)$ of a system $\mathcal{S}$ by applying the output $y(t)$ to a T second delay (Fig. 1.29a). If the system is time invariant, then the delayed output $y(t - T)$ can also be obtained by first delaying the input $x(t)$ before applying it to the system, as shown in Fig. 1.29b. In other words, the system $\mathcal{S}$ and the time delay commute if the system $\mathcal{S}$ is time invariant. This would not be true for time-varying systems. Consider, for instance, a time-varying system specified by $y(t) = e^{-t}x(t)$. The output for such a system in Fig. 1.29a is $e^{-(t-T)}x(t - T)$. In contrast, the output for the system in Fig. 1.29b is $e^{-t}x(t - T)$.

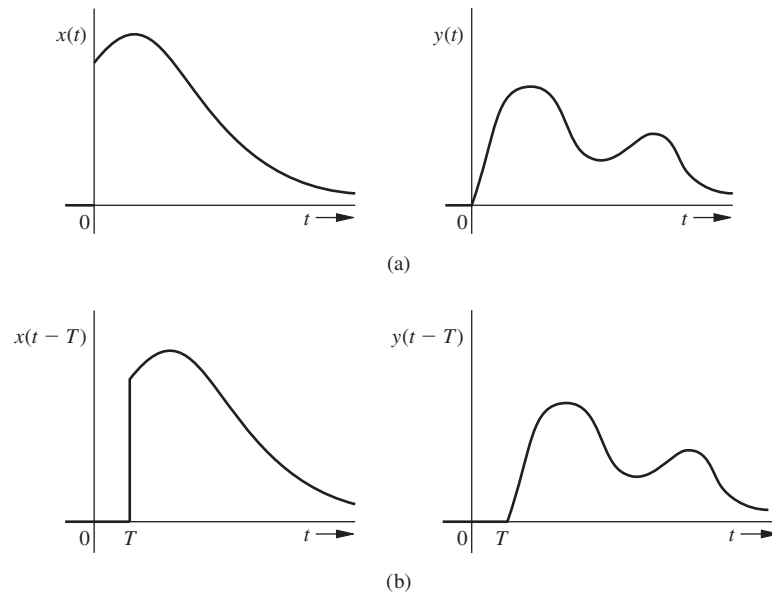


Figure 1.28 Time-invariance property.

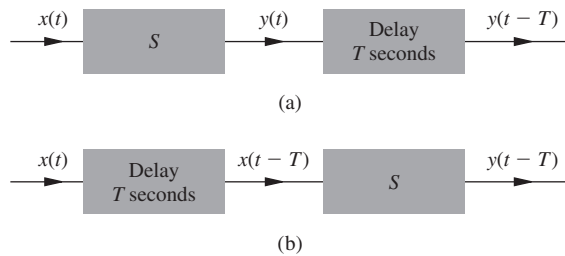


Figure 1.29 Illustration of time-invariance property.

It is possible to verify that the system in Fig. 1.26 is a time-invariant system. Networks composed of *RLC* elements and other commonly used active elements such as transistors are time-invariant systems. A system with an input–output relationship described by a linear differential equation of the form given in Ex. 1.10 [Eq. (1.25)] is a linear time-invariant (LTI) system when the coefficients a_i and b_i of such equation are constants. If these coefficients are functions of time, then the system is a linear *time-varying* system.

The system described in Drill 1.12 is linear time varying. Another familiar example of a time-varying system is the carbon microphone, in which the resistance R is a function of the mechanical pressure generated by sound waves on the carbon granules of the microphone. The output current from the microphone is thus modulated by the sound waves, as desired.

EXAMPLE 1.11 Assessing System Time Invariance

Determine the time invariance of the following systems: **(a)** $y(t) = x(t)u(t)$ and **(b)** $y(t) = \frac{d}{dt}x(t)$.

(a) In this case, the output equals the input for $t \geq 0$ and is otherwise zero. Clearly, the input is being modified by a time-dependent function, so the system is likely time variant. We can prove that the system is not time invariant through a counterexample. Letting $x_1(t) = \delta(t+1)$, we see that $y_1(t) = 0$. However, $x_2(t) = x_1(t-2) = \delta(t-1)$ produces an output of $y_2(t) = \delta(t-1)$, which does equal $y_1(t-2) = 0$ as time-invariance would require. Thus, $y(t) = x(t)u(t)$ is a time variant system.

(b) Although it appears that $x(t)$ is being modified by a time-dependent function, this is not the case. The output of this system is simply the slope of the input. If the input is delayed, so too is the output. Applying input $x(t)$ to the system produces output $y(t) = \frac{d}{dt}x(t)$; delaying this output by T produces $y(t-T) = \frac{d}{d(t-T)}x(t-T) = \frac{d}{dt}x(t-T)$. This is just the output of the system to a delayed input $x(t-T)$. Since the T -delayed output of the system to input $x(t)$ equals the output of the system to the T -delayed input $x(t-T)$, the system is time invariant.

DRILL 1.14 A Time-Variant System

Show that a system described by the following equation is a time-varying-parameter system:

$$y(t) = (\sin t)x(t-2)$$

[Hint: Show that the system fails to satisfy the time-invariance property.]

1.7-3 Instantaneous and Dynamic Systems

As observed earlier, a system's output at any instant t generally depends on the entire past input. However, in a special class of systems, the output at any instant t depends only on its input at that

instant. In resistive networks, for example, any output of the network at some instant t depends only on the input at the instant t . In these systems, past history is irrelevant in determining the response. Such systems are said to be *instantaneous* or *memoryless* systems. More precisely, a system is said to be instantaneous (or memoryless) if its output at any instant t depends, at most, on the strength of its input(s) at the same instant t , and not on any past or future values of the input(s). Otherwise, the system is said to be *dynamic* (or a system with memory). A system whose response at t is completely determined by the input signals over the past T seconds [interval from $(t - T)$ to t] is a *finite-memory system* with a memory of T seconds. Networks containing inductive and capacitive elements generally have infinite memory because the response of such networks at any instant t is determined by their inputs over the entire past $(-\infty, t)$. This is true for the RC circuit of Fig. 1.26.

EXAMPLE 1.12 Assessing System Memory

Determine whether the following systems are memoryless: **(a)** $y(t - 1) = 2x(t - 1)$, **(b)** $y(t) = \frac{d}{dt}x(t)$, and **(c)** $y(t) = (t - 1)x(t)$.

(a) In this case, the output at time $t - 1$ is just twice the input at the same time $t - 1$. Since the output at a particular time depends only on the strength of the input at the same time, the system is memoryless.

(b) Although it appears that the output $y(t)$ at time t depends on the input $x(t)$ at the same time t , we know that the slope (derivative) of $x(t)$ cannot be determined solely from a single point. There must be some memory, even if infinitesimally small, involved. This is confirmed by using the fundamental theorem of calculus to express the system as

$$y(t) = \lim_{T \rightarrow 0} \frac{x(t) - x(t - T)}{T}$$

Since the output at a particular time depends on more than just the input at the same time, the system is not memoryless.

(c) The output $y(t)$ at time t is just the input $x(t)$ at the same time t multiplied by the (time-dependent) coefficient $t - 1$. Since the output at a particular time depends only on the strength of the input at the same time, the system is memoryless.

1.7-4 Causal and Noncausal Systems

A *causal* (also known as a *physical* or *nonanticipative*) system is one for which the output at any instant t_0 depends only on the value of the input $x(t)$ for $t \leq t_0$. In other words, the value of the output at the present instant depends only on the past and present values of the input $x(t)$, not on its future values. To put it simply, in a causal system the output cannot start before the input is applied. If the response starts before the input, it means that the system knows the input in the

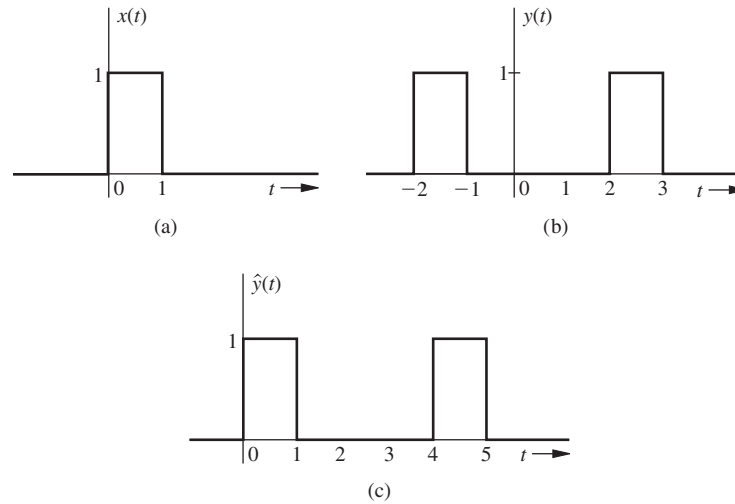


Figure 1.30 Input–output of a noncausal system and the causal output achieved by delay.

future and acts on this knowledge before the input is applied. A system that violates the condition of causality is called a *noncausal* (or *anticipative*) system.

Any practical system that operates in real time[†] must necessarily be causal. We do not yet know how to build a system that can respond to future inputs (inputs not yet applied). A noncausal system is a prophetic system that knows the future input and acts on it in the present. Thus, if we apply an input starting at $t = 0$ to a noncausal system, the output would begin even before $t = 0$. For example, consider the system specified by

$$y(t) = x(t - 2) + x(t + 2) \quad (1.26)$$

For the input $x(t)$ illustrated in Fig. 1.30a, the output $y(t)$, as computed from Eq. (1.26) (shown in Fig. 1.30b), starts even before the input is applied. Equation (1.26) shows that $y(t)$, the output at t , is given by the sum of the input values 2 seconds before and 2 seconds after t (at $t - 2$ and $t + 2$, respectively). But if we are operating the system in real time at t , we do not know what the value of the input will be 2 seconds later. Thus it is impossible to implement this system in real time. For this reason, noncausal systems are unrealizable in *real time*.

EXAMPLE 1.13 Assessing System Causality

Determine whether the following systems are causal: **(a)** $y(t) = x(-t)$, **(b)** $y(t) = x(t + 1)$, and **(c)** $y(t + 1) = x(t)$.

[†] In real-time operations, the response to an input is essentially simultaneous (contemporaneous) with the input itself.

(a) Here, the output is a reflection of the input. We can easily use a counterexample to disprove the causality of this system. The input $x(t) = \delta(t - 1)$, which is nonzero at $t = 1$, produces an output $y(t) = \delta(t + 1)$, which is nonzero at $t = -1$, a time 2 seconds earlier than the input! Clearly the system is not causal.

(b) In this case, the output at time t depends on the input at future time of $t + 1$. Clearly the system is not causal.

(c) In this case, the output at time $t + 1$ depends on the input one second in the past, at time t . Since the output does not depend on future values of the input, the system is causal.

WHY STUDY NONCAUSAL SYSTEMS?

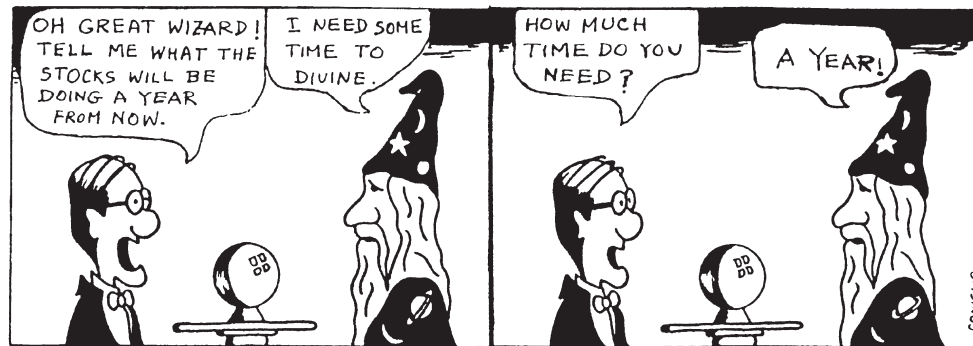
The foregoing discussion may suggest that noncausal systems have no practical purpose. This is not the case; they are valuable in the study of systems for several reasons. First, noncausal systems *are* realizable when the independent variable is other than “time” (e.g., *space*). Consider, for example, an electric charge of density $q(x)$ placed along the x axis for $x \geq 0$. This charge density produces an electric field $E(x)$ that is present at every point on the x axis from $x = -\infty$ to ∞ . In this case the input [i.e., the charge density $q(x)$] starts at $x = 0$, but its output [the electric field $E(x)$] begins before $x = 0$. Clearly, this space-charge system is noncausal. This discussion shows that only temporal systems (systems with time as independent variable) must be causal to be realizable. The terms “before” and “after” have a special connection to causality only when the independent variable is time. This connection is lost for variables other than time. Nontemporal systems, such as those occurring in optics, can be noncausal and still realizable.

Moreover, even for temporal systems, such as those used for signal processing, the study of noncausal systems is important. In such systems we may have all input data prerecorded. This often happens with speech, geophysical, and meteorological signals, and with space probes. In such cases, the input’s future values are available to us. For example, suppose we had a set of input signal records available for the system described by Eq. (1.26). We can then compute $y(t)$ since, for any t , we need only refer to the records to find the input’s value 2 seconds before and 2 seconds after t . Thus, noncausal systems can be realized, although not in real time. We may therefore be able to realize a noncausal system, provided we are willing to accept a time delay in the output. Consider a system whose output $\hat{y}(t)$ is the same as $y(t)$ in Eq. (1.26) delayed by 2 seconds (Fig. 1.30c), so that

$$\hat{y}(t) = y(t - 2) = x(t - 4) + x(t)$$

Here the value of the output $\hat{y}$ at any instant t is the sum of the values of the input x at t and at the instant 4 seconds earlier [at $(t - 4)$]. In this case, the output at any instant t does not depend on future values of the input, and the system is causal. The output of this system, which is $\hat{y}(t)$, is identical to that in Eq. (1.26) or Fig. 1.30b except for a delay of 2 seconds. Thus, a noncausal system may be realized or satisfactorily approximated in real time by using a causal system with a delay.

A third reason for studying noncausal systems is that they provide an upper bound on the performance of causal systems. For example, if we wish to design a filter for separating a signal from noise, then the optimum filter is invariably a noncausal system. Although unrealizable, this



Noncausal systems are realizable with time delay!

noncausal system's performance acts as the upper limit on what can be achieved and gives us a standard for evaluating the performance of causal filters.

At first glance, noncausal systems may seem to be inscrutable. Actually, there is nothing mysterious about these systems and their approximate realization through physical systems with delay. If we want to know what will happen one year from now, we have two choices: go to a prophet (an unrealizable person) who can give the answers instantly, or go to a wise man and allow him a delay of one year to give us the answer! If the wise man is truly wise, he may even be able, by studying trends, to shrewdly guess the future very closely with a delay of less than a year. Such is the case with noncausal systems—nothing more and nothing less.

DRILL 1.15 A Noncausal System

Show that a system described by the following equation is noncausal:

$$y(t) = \int_{t-5}^{t+5} x(\tau) d\tau$$

Show that this system can be realized physically if we accept a delay of 5 seconds in the output.

1.7-5 Continuous-Time and Discrete-Time Systems

Signals defined or specified over a continuous range of time are *continuous-time signals*, denoted by symbols $x(t)$, $y(t)$, and so on. Systems whose inputs and outputs are continuous-time signals are *continuous-time systems*. On the other hand, signals defined only at discrete instants of time $t_0, t_1, t_2, \dots, t_n, \dots$ are *discrete-time signals*, denoted by the symbols $x(t_n)$, $y(t_n)$, and so on, where n is some integer. Systems whose inputs and outputs are discrete-time signals are *discrete-time systems*. A digital computer is a familiar example of this type of system. In practice, discrete-time signals can arise from sampling continuous-time signals. For example, when the sampling is

uniform, the discrete instants $t_0, t_1, t_2, \dots$ are uniformly spaced so that

$$t_{k+1} - t_k = T \quad \text{for all } k$$

In such case, the discrete-time signals represented by the samples of continuous-time signals $x(t), y(t)$, and so on can be expressed as $x(nT), y(nT)$, and so on; for convenience, we further simplify this notation to $x[n], y[n], \dots$, where it is understood that $x[n] = x(nT)$ and that n is some integer. A typical discrete-time signal is shown in Fig. 1.31. A discrete-time signal may also be viewed as a sequence of numbers $\dots, x[-1], x[0], x[1], x[2], \dots$. Thus, a discrete-time system may be seen as processing a sequence of numbers $x[n]$ and yielding as an output another sequence of numbers $y[n]$.

Discrete-time signals arise naturally in situations that are inherently discrete time, such as population studies, amortization problems, national income models, and radar tracking. They may also arise as a result of sampling continuous-time signals in sampled data systems, digital filtering, and the like. Digital filtering is a particularly interesting application in which continuous-time signals are processed by using discrete-time systems, as shown in Fig. 1.32. A continuous-time signal $x(t)$ is first sampled to convert it into a discrete-time signal $x[n]$, which then is processed by the discrete-time system to yield a discrete-time output $y[n]$. A continuous-time signal $y(t)$ is finally constructed from $y[n]$. In this manner, we can process a continuous-time signal with an appropriate discrete-time system such as a digital computer. Because discrete-time systems have several significant advantages over continuous-time systems, there is an accelerating trend toward processing continuous-time signals with discrete-time systems.

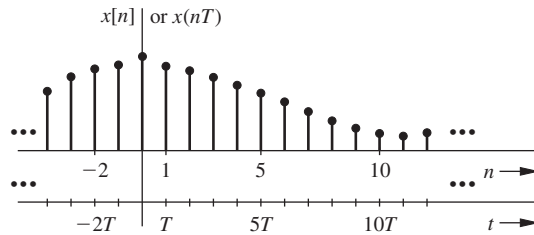


Figure 1.31 A discrete-time signal.

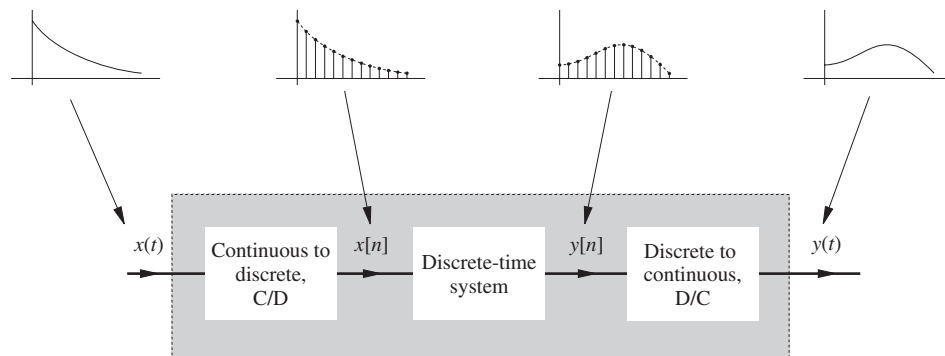


Figure 1.32 Processing continuous-time signals by discrete-time systems.

1.7-6 Analog and Digital Systems

Analog and digital signals are discussed in Sec. 1.3-2. A system whose input and output signals are analog is an *analog system*; a system whose input and output signals are digital is a *digital system*. A digital computer is an example of a digital (binary) system. Observe that a digital computer is a digital as well as a discrete-time system.

1.7-7 Invertible and Noninvertible Systems

A system $\mathcal{S}$ performs certain operation(s) on input signal(s). If we can obtain the input $x(t)$ back from the corresponding output $y(t)$ by some operation, the system $\mathcal{S}$ is said to be *invertible*. When several different inputs result in the same output (as in a rectifier), it is impossible to obtain the input from the output, and the system is *noninvertible*. Therefore, for an invertible system, it is essential that every input have a unique output so that there is a one-to-one mapping between an input and the corresponding output. The system that achieves the inverse operation [of obtaining $x(t)$ from $y(t)$] is the *inverse system* for $\mathcal{S}$. For instance, if $\mathcal{S}$ is an ideal integrator, then its inverse system is an ideal differentiator. Consider a system $\mathcal{S}$ connected in tandem with its inverse $\mathcal{S}_i$, as shown in Fig. 1.33. The input $x(t)$ to this tandem system results in signal $y(t)$ at the output of $\mathcal{S}$, and the signal $y(t)$, which now acts as an input to $\mathcal{S}_i$, yields back the signal $x(t)$ at the output of $\mathcal{S}_i$. Thus, $\mathcal{S}_i$ undoes the operation of $\mathcal{S}$ on $x(t)$, yielding back $x(t)$. A system whose output is equal to the input (for all possible inputs) is an *identity system*. Cascading a system with its inverse system, as shown in Fig. 1.33, results in an identity system.

In contrast, a rectifier, specified by an equation $y(t) = |x(t)|$, is noninvertible because the rectification operation cannot be undone.

Inverse systems are very important in signal processing. In many applications, the signals are distorted during the processing, and it is necessary to undo the distortion. For instance, in transmission of data over a communication channel, the signals are distorted owing to non-ideal frequency response and finite bandwidth of a channel. It is necessary to restore the signal as closely as possible to its original shape. Such equalization is also used in audio systems and photographic systems.

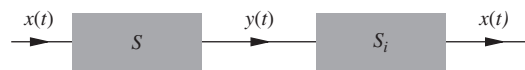


Figure 1.33 A cascade of a system with its inverse results in an identity system.

EXAMPLE 1.14 Assessing System Invertibility

Determine whether the following systems are invertible: **(a)** $y(t) = x(-t)$, **(b)** $y(t) = tx(t)$, and **(c)** $y(t) = \frac{d}{dt}x(t)$.

(a) Here, the output is a reflection of the input, which does not cause any loss to the input. The input can, in fact, be exactly recovered by simply reflecting the output [$x(t) = y(-t)$], which is to say that a reflecting system is its own inverse. Thus, $y(t) = x(-t)$ is an invertible system.

(b) In this case, one might be tempted to recover the input from the output as $x(t) = \frac{1}{t}y(t)$. This approach works almost everywhere, except at $t = 0$ where the input value $x(0)$ cannot be recovered. Due to this single lost point, the system $y(t) = tx(t)$ is not invertible.

(c) Differentiation eliminates any dc component. For example, the inputs $x_1(t) = 1$ and $x_2(t) = 2$ both produce the same output $y(t) = 0$. Given only $y(t) = 0$, it is impossible to know if the original input was $x_1(t) = 1$, $x_2(t) = 2$, or something else entirely. Since unique inputs do produce unique outputs, we know that $y(t) = \frac{d}{dt}x(t)$ is not an invertible system.

1.7-8 Stable and Unstable Systems

Systems can also be classified as *stable* or *unstable* systems. Stability can be *internal* or *external*. If every *bounded input* applied at the input terminal results in a *bounded output*, the system is said to be stable *externally*. External stability can be ascertained by measurements at the external terminals (input and output) of the system. This type of stability is also known as the stability in the BIBO (bounded-input/bounded-output) sense. The concept of internal stability is postponed to Ch. 2 because it requires some understanding of internal system behavior, introduced in that chapter.

EXAMPLE 1.15 Assessing System BIBO Stability

Determine whether the following systems are BIBO-stable: (a) $y(t) = x^2(t)$, (b) $y(t) = tx(t)$, and (c) $y(t) = \frac{d}{dt}x(t)$.

(a) This system squares an input to produce the output. If the input is bounded, which is to say that $|x(t)| \leq M_x < \infty$ for all t , then we see that

$$|y(t)| = |x^2(t)| = |x(t)|^2 \leq M_x^2 < \infty$$

Since the output amplitude is guaranteed to be bounded for any bounded-amplitude input, the system $y(t) = x^2(t)$ is BIBO-stable.

(b) We can prove that $y(t) = tx(t)$ is not BIBO-stable with a simple example. The bounded-amplitude input $x(t) = u(t)$ produces the output $y(t) = tu(t)$ whose amplitude grows to infinity as $t \rightarrow \infty$. Thus, $y(t) = tx(t)$ is a BIBO-unstable system.

(c) We can prove that $y(t) = \frac{d}{dt}x(t)$ is not BIBO-stable with an example. The bounded-amplitude input $x(t) = u(t)$ produces the output $y(t) = \delta(t)$ whose amplitude is infinite at $t = 0$. Thus, $y(t) = \frac{d}{dt}x(t)$ is a BIBO-unstable system.

DRILL 1.16 A Noninvertible BIBO-Stable System

Show that a system described by the equation $y(t) = x^2(t)$ is noninvertible but BIBO-stable.

1.8 SYSTEM MODEL: INPUT–OUTPUT DESCRIPTION

A system description in terms of the measurements at the input and output terminals is called the *input–output description*. As mentioned earlier, systems theory encompasses a variety of systems, such as electrical, mechanical, hydraulic, acoustic, electromechanical, and chemical, as well as social, political, economic, and biological. The first step in analyzing any system is the construction of a system model, which is a mathematical expression or a rule that satisfactorily approximates the dynamical behavior of the system. In this chapter we shall consider only continuous-time systems. Modeling of discrete-time systems is discussed in Ch. 3.

1.8-1 Electrical Systems

To construct a system model, we must study the relationships between different variables in the system. In electrical systems, for example, we must determine a satisfactory model for the voltage–current relationship of each element, such as Ohm’s law for a resistor. In addition, we must determine the various constraints on voltages and currents when several electrical elements are interconnected. These are the laws of interconnection—the well-known Kirchhoff laws for voltage and current (KVL and KCL). From all these equations, we eliminate unwanted variables to obtain equation(s) relating the desired output variable(s) to the input(s). The following examples demonstrate the procedure of deriving input–output relationships for some LTI electrical systems.

EXAMPLE 1.16 Input–Output Equation of a Series RLC Circuit

For the series RLC circuit of Fig. 1.34, find the input–output equation relating the input voltage $x(t)$ to the output current (loop current) $y(t)$.

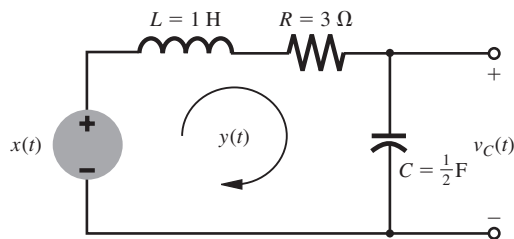


Figure 1.34 Circuit for Ex. 1.16.

Application of Kirchhoff’s voltage law around the loop yields

$$v_L(t) + v_R(t) + v_C(t) = x(t)$$

By using the voltage-current laws of each element (inductor, resistor, and capacitor), we can express this equation as

$$\frac{dy(t)}{dt} + 3y(t) + 2 \int_{-\infty}^t y(\tau) d\tau = x(t) \quad (1.27)$$

Differentiating both sides of this equation, we obtain

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} \quad (1.28)$$

This differential equation is the input–output relationship between the output $y(t)$ and the input $x(t)$.

It proves convenient to use a compact notation D for the differential operator d/dt . This notation can be repeatedly applied. Thus,

$$\frac{dy(t)}{dt} \equiv Dy(t), \quad \frac{d^2y(t)}{dt^2} \equiv D^2y(t), \quad \dots, \quad \frac{d^Ny(t)}{dt^N} \equiv D^Ny(t)$$

With this notation, Eq. (1.28) can be expressed as

$$(D^2 + 3D + 2)y(t) = Dx(t) \quad (1.29)$$

The differential operator is the inverse of the integral operator, so we can use the operator $1/D$ to represent integration.[†]

$$\int_{-\infty}^t y(\tau) d\tau \equiv \frac{1}{D}y(t)$$

[†] Use of operator $1/D$ for integration generates some subtle mathematical difficulties because the operators D and $1/D$ do not commute. For instance, we know that $D(1/D) = 1$ because

$$\frac{d}{dt} \left[\int_{-\infty}^t y(\tau) d\tau \right] = y(t)$$

However, $(1/D)D$ is not necessarily unity. Use of Cramer's rule in solving simultaneous integro-differential equations will always result in cancellation of operators $1/D$ and D . This procedure may yield erroneous results when the factor D occurs in the numerator as well as in the denominator. This happens, for instance, in circuits with all-inductor loops or all-capacitor cut sets. To eliminate this problem, avoid the integral operation in system equations so that the resulting equations are differential rather than integro-differential. In electrical circuits, this can be done by using charge (instead of current) variables in loops containing capacitors and choosing current variables for loops without capacitors. In the literature this problem of commutativity of D and $1/D$ is largely ignored. As mentioned earlier, such a procedure gives erroneous results only in special systems, such as the circuits with all-inductor loops or all-capacitor cut sets. Fortunately such systems constitute a very small fraction of the systems we deal with. For further discussion of this topic and a correct method of handling problems involving integrals, see [4].

Consequently, Eq. (1.27) can be expressed as

$$\left(D + 3 + \frac{2}{D}\right)y(t) = x(t)$$

Multiplying both sides by D to differentiate the expression, we obtain

$$(D^2 + 3D + 2)y(t) = Dx(t)$$

which is identical to Eq. (1.29).

Recall that Eq. (1.29) is not an algebraic equation, and $D^2 + 3D + 2$ is not an algebraic term that multiplies $y(t)$; it is an operator that operates on $y(t)$. It means that we must perform the following operations on $y(t)$: take the second derivative of $y(t)$ and add to it 3 times the first derivative of $y(t)$ and 2 times $y(t)$. Clearly, a polynomial in D multiplied by $y(t)$ represents a certain differential operation on $y(t)$.

EXAMPLE 1.17 Input–Output Equation of a Series RC Circuit

Using operator notation, find the equation relating input to output for the series RC circuit of Fig. 1.35 if the input is the voltage $x(t)$ and output is

- (a) the loop current $i(t)$
- (b) the capacitor voltage $y(t)$

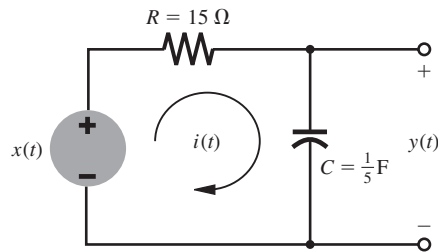


Figure 1.35 Circuit for Ex. 1.17

- (a) The loop equation for the circuit is

$$Ri(t) + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = x(t)$$

or

$$15i(t) + 5 \int_{-\infty}^t i(\tau) d\tau = x(t)$$

With operator notation, this equation can be expressed as

$$15i(t) + \frac{5}{D}i(t) = x(t) \quad (1.30)$$

(b) Multiplying both sides of Eq. (1.30) by D (i.e., differentiating the equation), we obtain

$$(15D + 5)i(t) = Dx(t)$$

Using the fact that $i(t) = C \frac{dy(t)}{dt} = \frac{1}{5}Dy(t)$, simple substitution yields

$$(3D + 1)y(t) = x(t) \quad (1.31)$$

DRILL 1.17 Input–Output Equation of a Series RLC Circuit with Inductor Voltage as Output

If the inductor voltage $v_L(t)$ is taken as the output, show that the RLC circuit in Fig. 1.34 has an input–output equation of $(D^2 + 3D + 2)v_L(t) = D^2x(t)$.

DRILL 1.18 Input–Output Equation of a Series RC Circuit with Capacitor Voltage as Output

If the capacitor voltage $v_C(t)$ is taken as the output, show that the RLC circuit in Fig. 1.34 has an input–output equation of $(D^2 + 3D + 2)v_C(t) = 2x(t)$.

1.8-2 Mechanical Systems

Planar motion can be resolved into translational (rectilinear) motion and rotational (torsional) motion. Translational motion will be considered first. We shall restrict ourselves to motions in one dimension.

TRANSLATIONAL SYSTEMS

The basic elements used in modeling translational systems are ideal masses, linear springs, and dashpots providing viscous damping. The laws of various mechanical elements are now discussed.

For a *mass* M (Fig. 1.36a), a force $x(t)$ causes a motion $y(t)$ and acceleration $\ddot{y}(t)$. From Newton's law of motion,

$$x(t) = M\ddot{y}(t) = M \frac{d^2y(t)}{dt^2} = MD^2y(t)$$

The force $x(t)$ required to stretch (or compress) a *linear spring* (Fig. 1.36b) by an amount $y(t)$ is given by

$$x(t) = Ky(t)$$

where K is the *stiffness* of the spring.

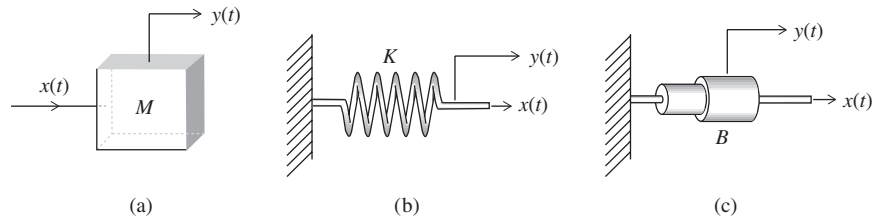


Figure 1.36 Some elements in translational mechanical systems.

For a *linear dashpot* (Fig. 1.36c), which operates by virtue of viscous friction, the force moving the dashpot is proportional to the relative velocity $\dot{y}(t)$ of one surface with respect to the other. Thus

$$x(t) = B\dot{y}(t) = B \frac{dy(t)}{dt} = BD\dot{y}(t)$$

where B is the *damping coefficient* of the dashpot or the viscous friction.

EXAMPLE 1.18 Input–Output Equation for a Translational Mechanical System

Find the input–output relationship for the translational mechanical system shown in Fig. 1.37a or its equivalent in Fig. 1.37b. The input is the force $x(t)$, and the output is the mass position $y(t)$.

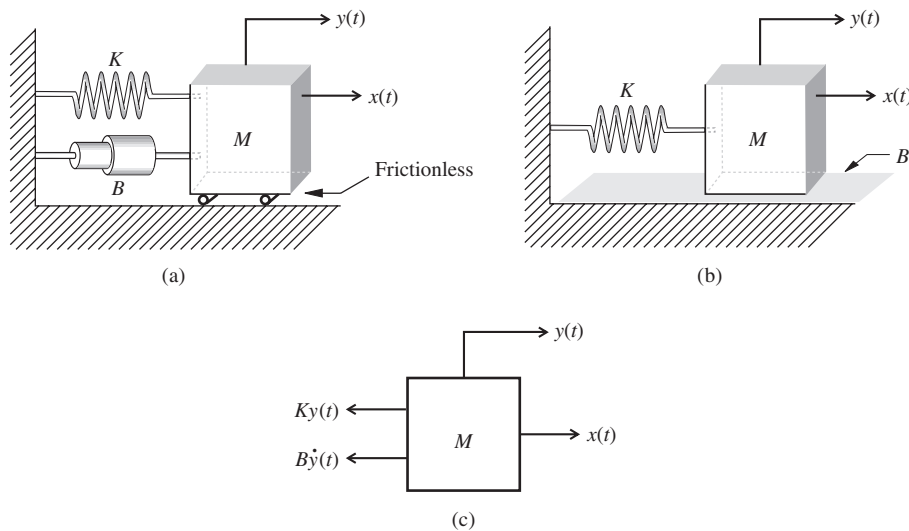


Figure 1.37 Mechanical system for Ex. 1.18

In mechanical systems it is helpful to draw a free-body diagram of each junction, which is a point at which two or more elements are connected. In Fig. 1.37, the point representing the mass is a junction. The displacement of the mass is denoted by $y(t)$. The spring is also stretched by the amount $y(t)$, and therefore it exerts a force $-Ky(t)$ on the mass. The dashpot exerts a force $-B\dot{y}(t)$ on the mass, as shown in the free-body diagram (Fig. 1.37c). By Newton's second law, the net force must be $M\ddot{y}(t)$. Therefore,

$$M\ddot{y}(t) = -B\dot{y}(t) - Ky(t) + x(t)$$

or

$$(MD^2 + BD + K)y(t) = x(t)$$

ROTATIONAL SYSTEMS

In rotational systems, the motion of a body may be defined as its motion about a certain axis. The variables used to describe rotational motion are torque (in place of force), angular position (in place of linear position), angular velocity (in place of linear velocity), and angular acceleration (in place of linear acceleration). The system elements are *rotational mass* or *moment of inertia* (in place of mass) and *torsional springs* and *torsional dashpots* (in place of linear springs and dashpots). The terminal equations for these elements are analogous to the corresponding equations for translational elements. If J is the moment of inertia (or rotational mass) of a rotating body about a certain axis, then the external torque required for this motion is equal to J (rotational mass) times the angular acceleration. If $\theta(t)$ is the angular position of the body, $\ddot{\theta}(t)$ is its angular acceleration, and

$$\text{torque} = J\ddot{\theta}(t) = J \frac{d^2\theta(t)}{dt^2} = JD^2\theta(t)$$

Similarly, if K is the stiffness of a torsional spring (per unit angular twist), and θ is the angular displacement of one terminal of the spring with respect to the other, then

$$\text{torque} = K\theta(t)$$

Finally, the torque due to viscous damping of a torsional dashpot with damping coefficient B is

$$\text{torque} = B\dot{\theta}(t) = BD\dot{\theta}(t)$$

EXAMPLE 1.19 Input–Output Equation for Aircraft Roll Angle

The attitude of an aircraft can be controlled by three sets of surfaces (shown shaded in Fig. 1.38): elevators, rudder, and ailerons. By manipulating these surfaces, one can set the aircraft on a desired flight path. The roll angle $\varphi(t)$ can be controlled by deflecting in the opposite direction the two aileron surfaces as shown in Fig. 1.38. Assuming only rolling motion, find the equation relating the roll angle $\varphi(t)$ to the input (deflection) $\theta(t)$.

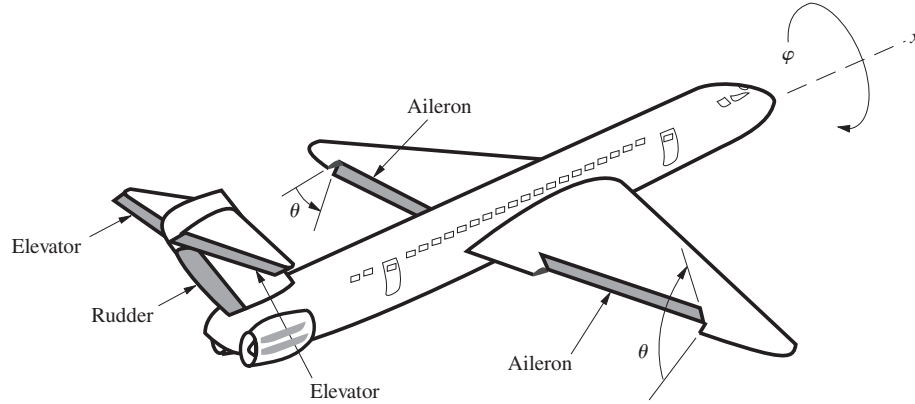


Figure 1.38 Attitude control of an airplane.

The aileron surfaces generate a torque about the roll axis proportional to the aileron deflection angle $\theta(t)$. Let this torque be $c\theta(t)$, where c is the constant of proportionality. Air friction dissipates the torque $B\dot{\phi}(t)$. The torque available for rolling motion is then $c\theta(t) - B\dot{\phi}(t)$. If J is the moment of inertia of the plane about the x axis (roll axis), then

$$\text{net torque} = J\ddot{\phi}(t) = c\theta(t) - B\dot{\phi}(t)$$

and

$$J\frac{d^2\phi(t)}{dt^2} + B\frac{d\phi(t)}{dt} = c\theta(t) \quad \text{or} \quad (JD^2 + BD)\phi(t) = c\theta(t)$$

This is the desired equation relating the output (roll angle $\phi(t)$) to the input (aileron angle $\theta(t)$).

The roll velocity $\omega(t)$ is $\dot{\phi}(t)$. If the desired output is the roll velocity $\omega(t)$ rather than the roll angle $\phi(t)$, then the input–output equation would be

$$J\frac{d\omega(t)}{dt} + B\omega(t) = c\theta(t) \quad \text{or} \quad (JD + B)\omega(t) = c\theta(t)$$

DRILL 1.19 Input–Output Equation of a Rotational Mechanical System

Torque $\mathcal{T}(t)$ is applied to the rotational mechanical system shown in Fig. 1.39a. The torsional spring stiffness is K ; the rotational mass (the cylinder's moment of inertia about the shaft) is J ; the viscous damping coefficient between the cylinder and the ground is B . Find the equation

relating the output angle $\theta(t)$ to the input torque $\mathcal{T}(t)$. [Hint: A free-body diagram is shown in Fig. 1.39b.]

ANSWER

$$J \frac{d^2\theta(t)}{dt^2} + B \frac{d\theta(t)}{dt} + K\theta(t) = \mathcal{T}(t)$$

or

$$(JD^2 + BD + K)\theta(t) = \mathcal{T}(t)$$

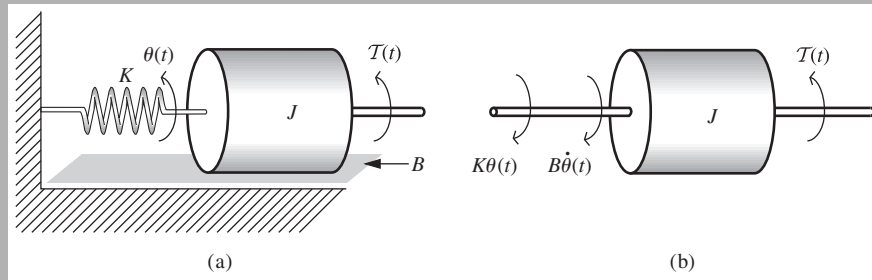


Figure 1.39 Rotational system for Drill 1.19.

1.8-3 Electromechanical Systems

A wide variety of electromechanical systems is used to convert electrical signals into mechanical motion (mechanical energy) and vice versa. Here we consider a rather simple example of an armature-controlled dc motor driven by a current source $x(t)$, as shown in Fig. 1.40a. The torque $\mathcal{T}(t)$ generated in the motor is proportional to the armature current $x(t)$. Therefore,

$$\mathcal{T}(t) = K_T x(t)$$

where K_T is a constant of the motor. This torque drives a mechanical load whose free-body diagram is shown in Fig. 1.40b. The viscous damping (with coefficient B) dissipates a torque $B\dot{\theta}(t)$. If J is the moment of inertia of the load (including the rotor of the motor), then the net torque $\mathcal{T}(t) - B\dot{\theta}(t)$ must be equal to $J\ddot{\theta}(t)$:

$$J\ddot{\theta}(t) = \mathcal{T}(t) - B\dot{\theta}(t)$$

Thus,

$$(JD^2 + BD)\theta(t) = \mathcal{T}(t) = K_T x(t)$$

which in conventional form can be expressed as

$$J \frac{d^2\theta(t)}{dt^2} + B \frac{d\theta(t)}{dt} = K_T x(t) \quad (1.32)$$